Medical Physics

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- **School of Physics**

Chapter 8:

Electromagnetic Induction

Chapter 8： Electromagnetic Induction

You will learn

1. The experimental evidence that a changing magnetic field induces an emf
2. How Faraday's law relates the induced emf in a loop to the change in magnetic flux through
3. How to determine the direction of an induced emf.
4. How to calculate the emf induced in a conductor moving through a magnetic field
5. How a changing magnetic flux generates an electric field that is very different from that produced by an arrangement of charges.
6. How to calculate the inductance and related magnetic energy
7. Know what is Ferromagnetism, paramagnetism, diamagnetism, magnetic domain, curie temperature, hysteresis loop
8. The expression of the electromagnetic Waves and travel with the speed of light.

Chapter 8： Electromagnetic Induction

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

$$\varepsilon = \int_b^a E_n \bullet dl =$$

$$\int_b^a v \times B \bullet dl$$

$$V_{ab} = El = vBl$$

$$L = \frac{N\Phi}{I} = \frac{\Psi}{I}$$

$$\varepsilon = -L\frac{dI}{dt}$$

Magnetic energy

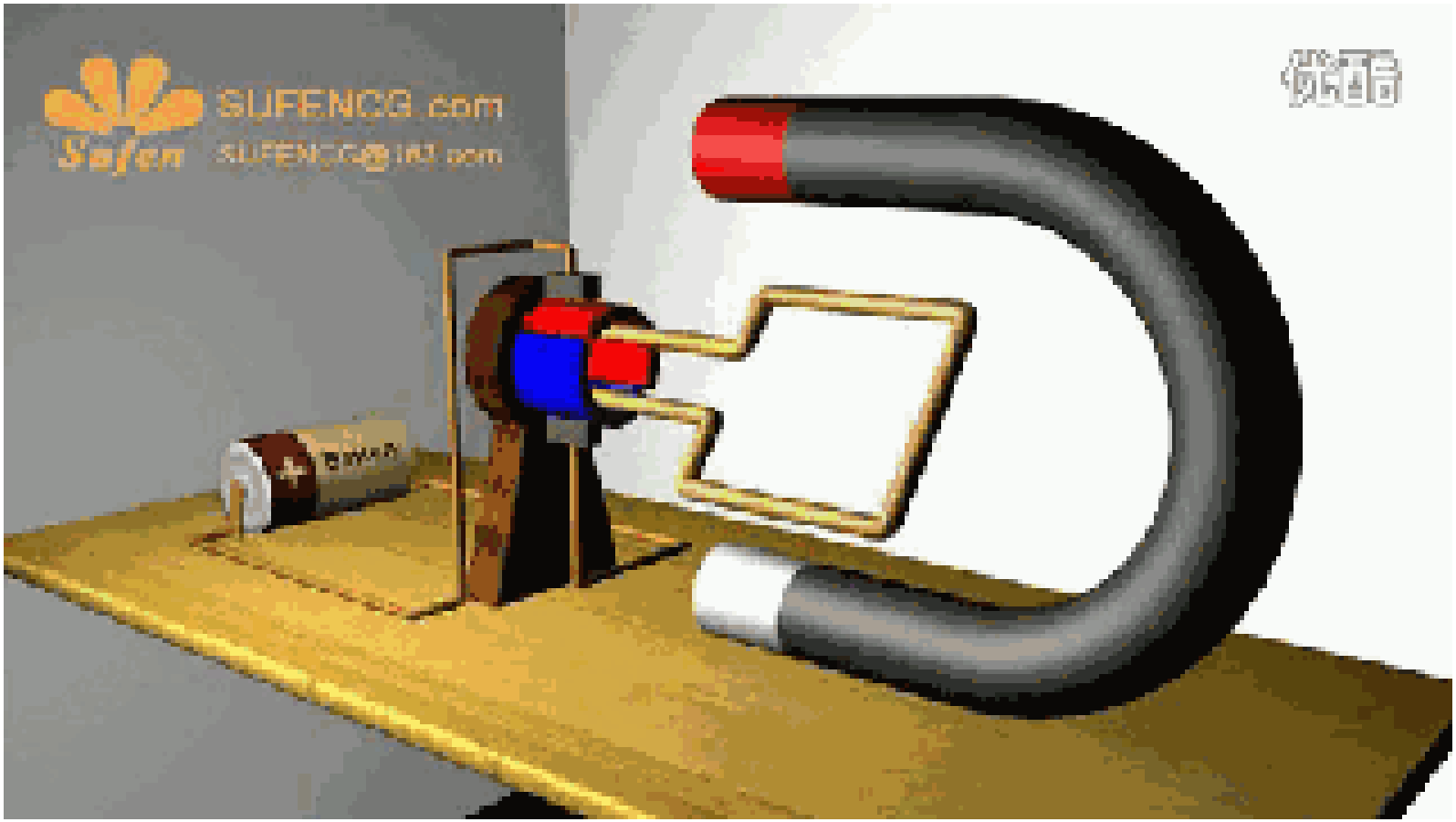
$$U = \frac{1}{2}LI^2$$

Electromagnetic Waves

$$E_x = E_0 \sin[\frac{2\pi}{\lambda}(z - ct)]$$

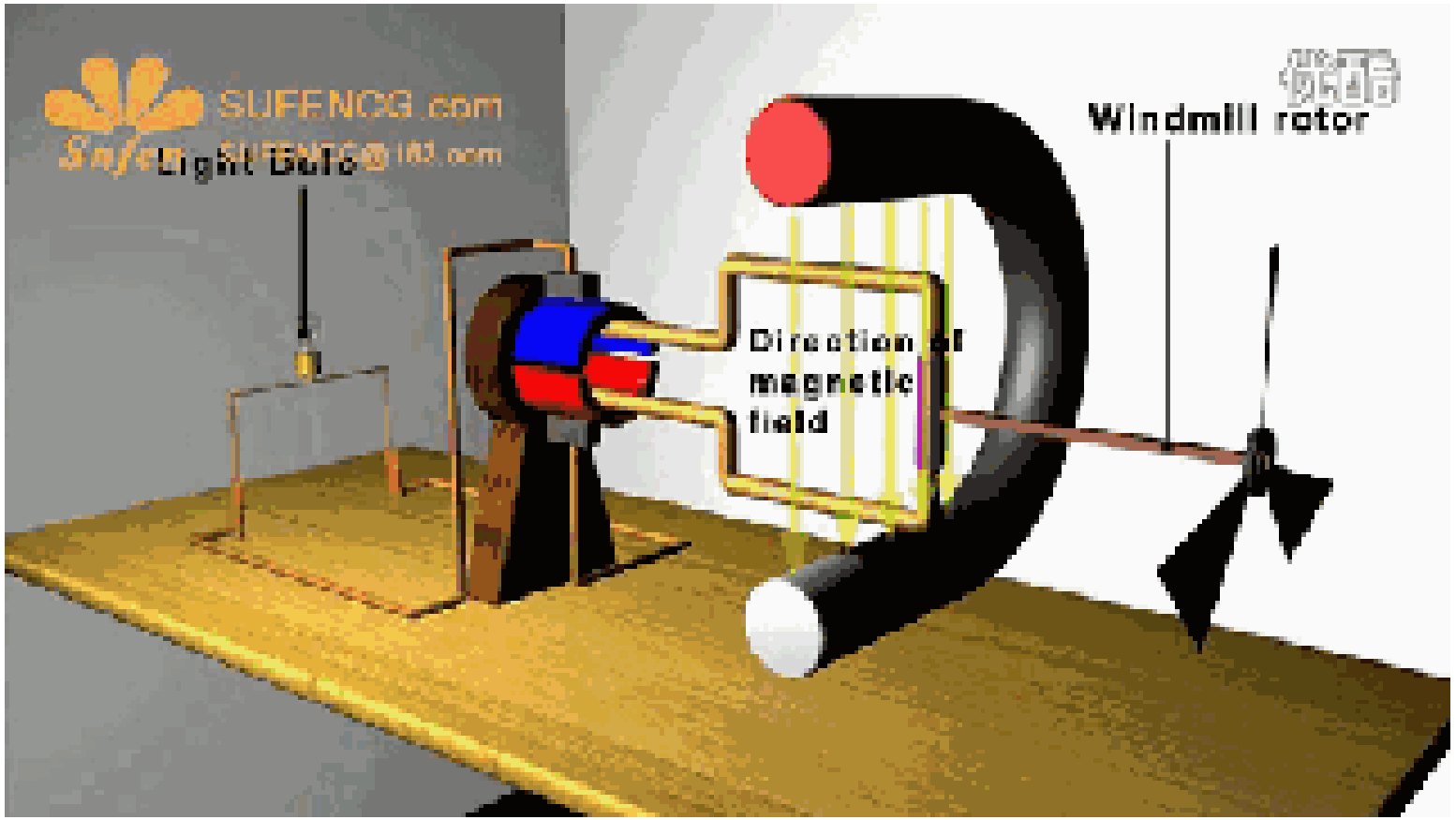
$$B_x = B_0 \sin[\frac{2\pi}{\lambda}(z - ct)]$$

Chapter 8： Electromagnetic Induction



Motor: electrical energy convert to kinetic energy

Electric generator



Electric generator: Kinetic energy convert to electrical energy

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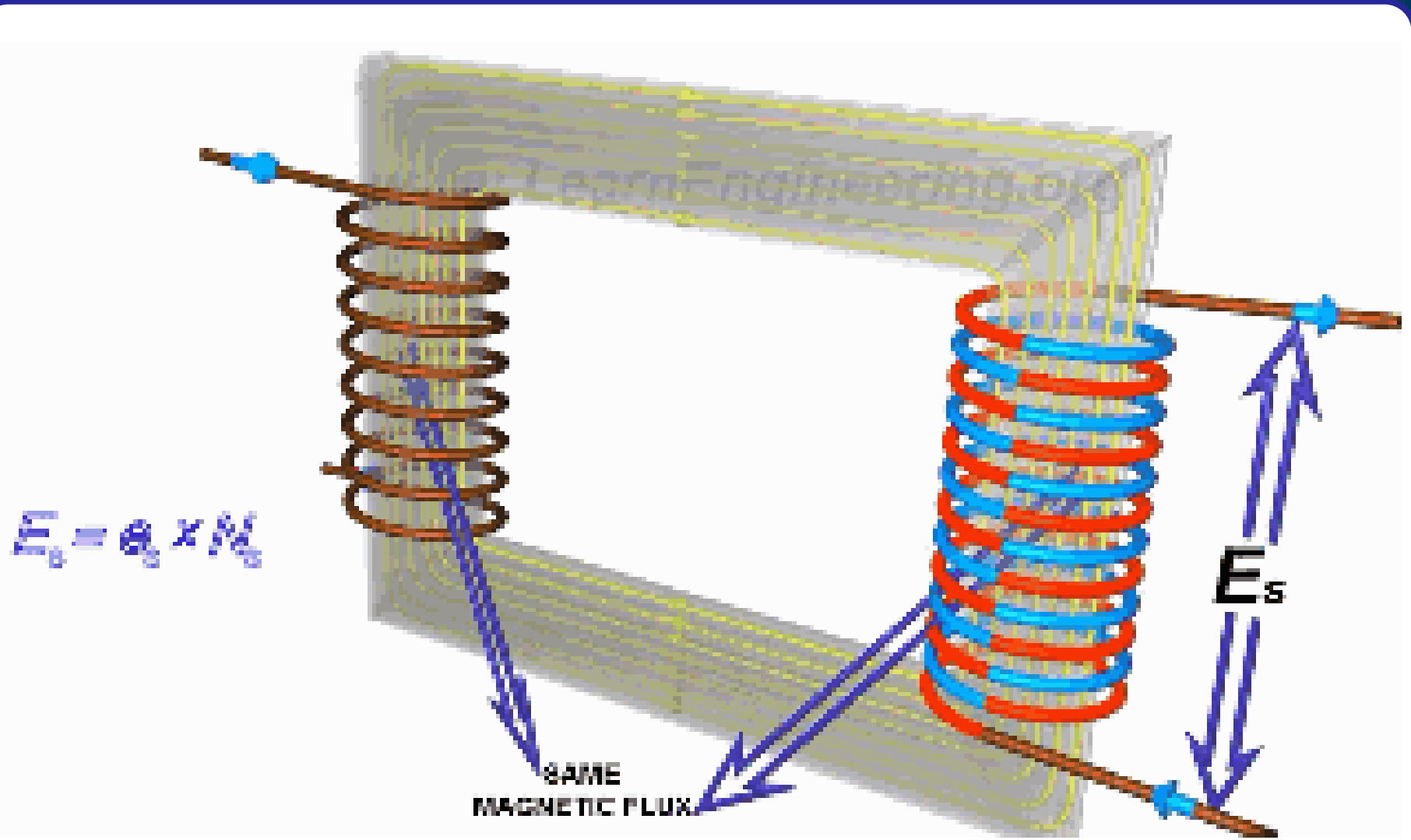
Hydropower station



Wireless charge technology



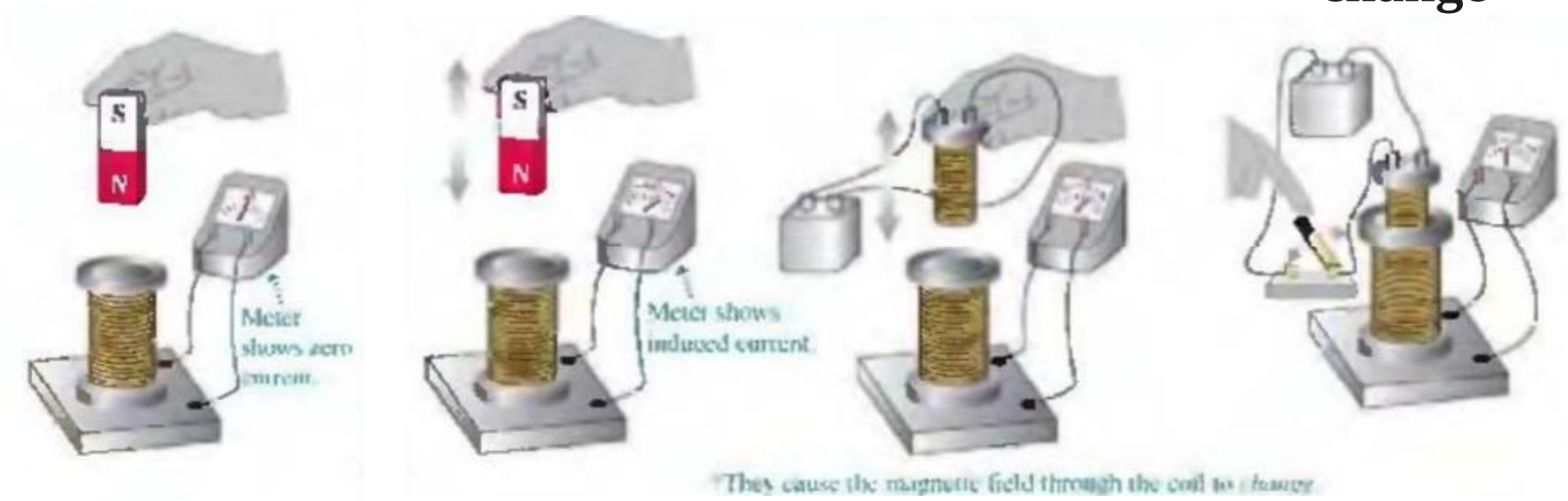
Electric transformator



8.1.1 Induced current

How to generate the induced current/electromotive-force/potential

Magnetic bar No move	Magnetic bar Move	Solenoid Move	Solenoid Current change



8.1.1 Faraday's law

Faraday's law of induction

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

ε: induced emf (unit: volt, V)

Φ_B: magnetic flux (unit: T m^2 or weber (Wb)

t: time

$$\Phi_B = \int B \cdot dA$$



Michael Faraday
（1791-1867）

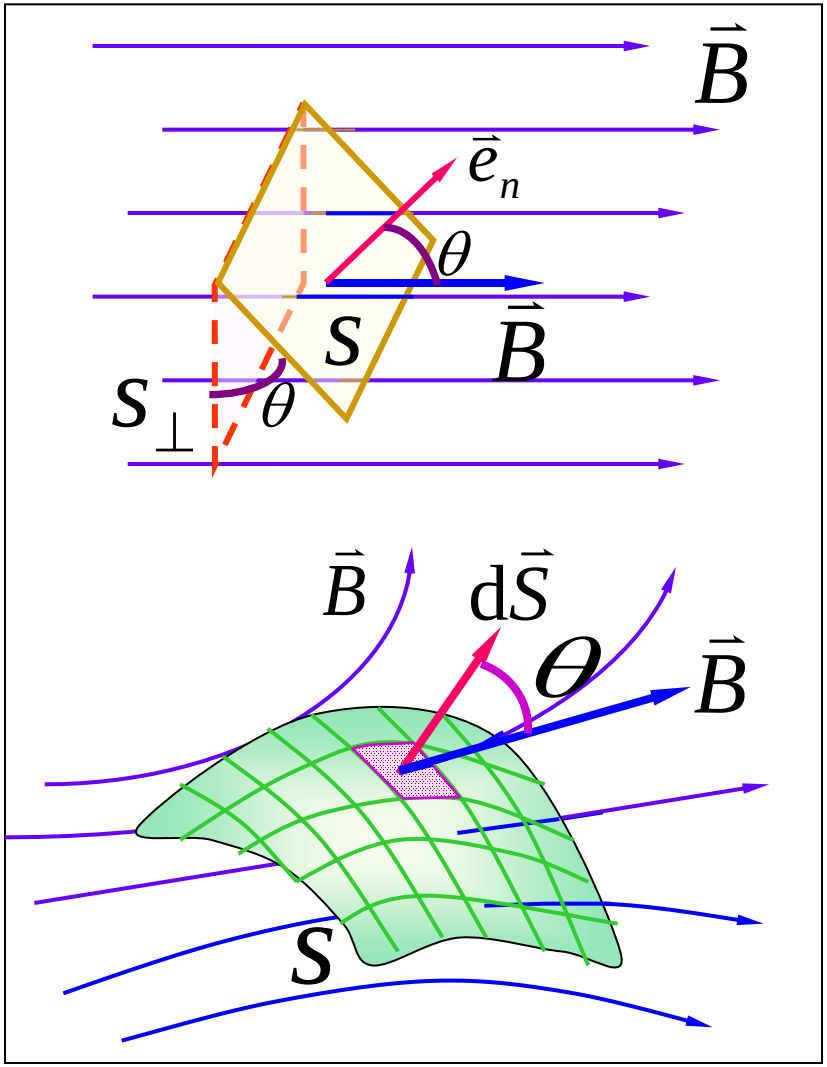
8.1.1 How to calculate the magnetic flux

For any unit area,
the magnetic field is

$$d\Phi = B\cos\theta dS = B_n dS$$

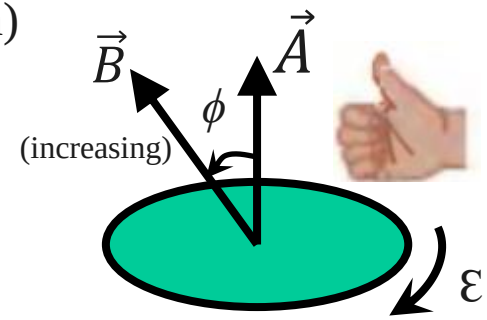
General case

$$\Phi = \int_S \vec{B}\cdot d\vec{S}$$



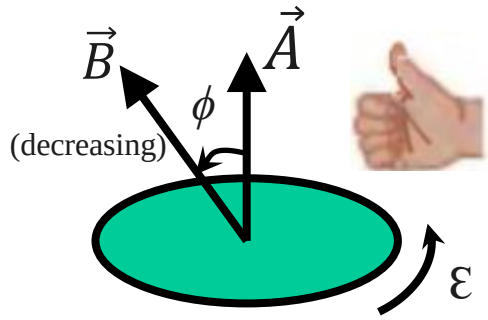
8.1.1 Direction of the induced emf

(a)



- Flux is positive $\Phi_B>0$
- $d\Phi_B/dt>0$
- Induced emf is negative $\mathcal{E}<0$.

(b)

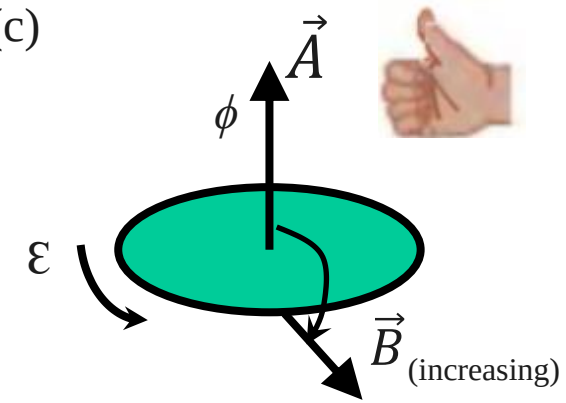


- When $\Phi_B>0$
- $d\Phi_B/dt<0$
- Induced emf is positive $\mathcal{E}>0$.

Step 1: Define a positive direction for the vector area A.
Step 2: From the directions of *A* and the magnetic field *B*, determine the sign of the magnetic flux ϕB and *its* rate of change.
Step 3: Determine the sign of the induced emf. If the flux is increasing, so $d\phi B/dt$ is positive, then the induced emf or current is negative; if the flux is decreasing, $d\phi B/dt$ is negative and the induced emf or current is positive.

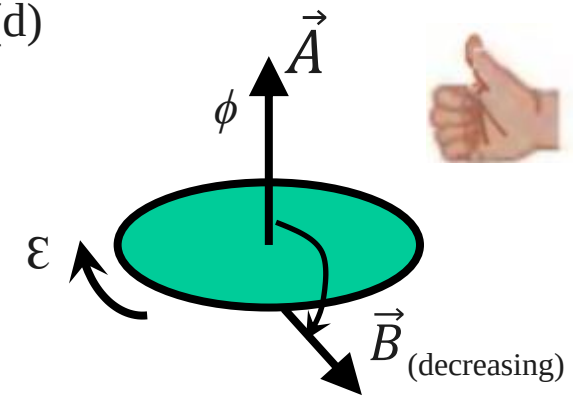
8.1.1 Direction of the induced emf

(c)



- When $\Phi_B<0$
- $d\Phi_B/dt<0$
- Induced emf is positive $\varepsilon>0$.

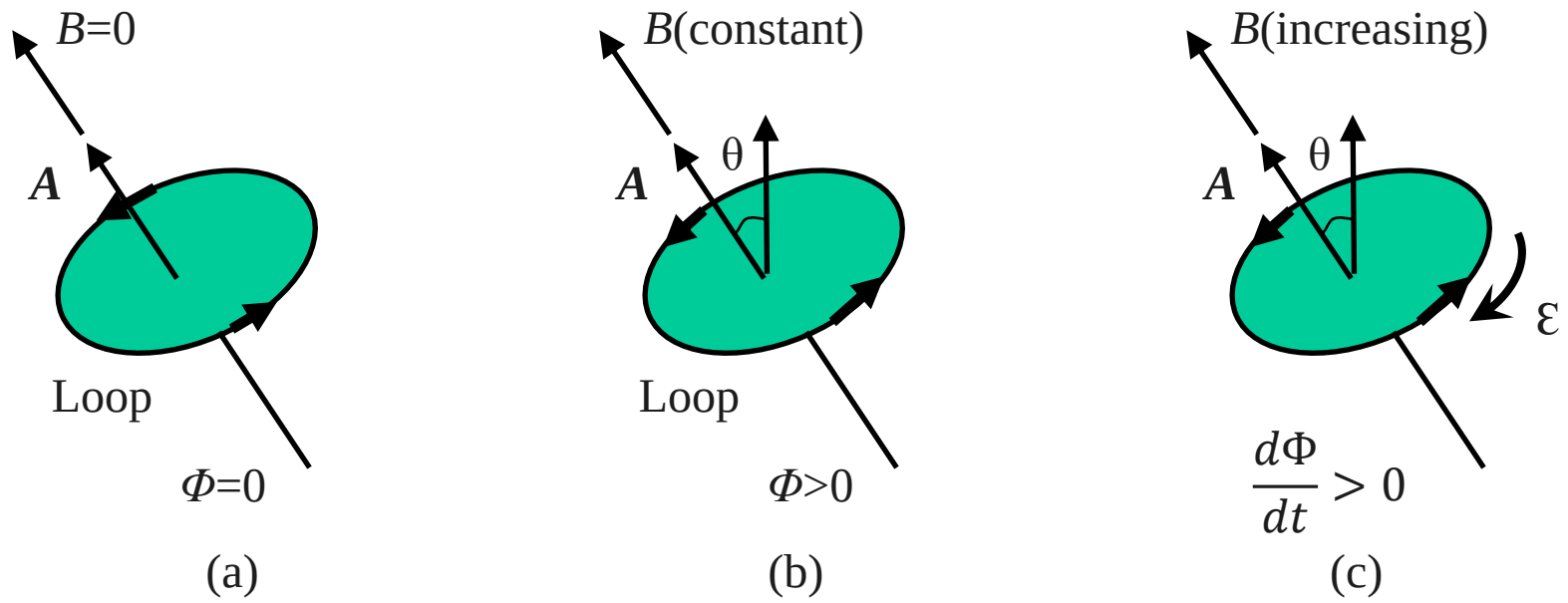
(d)



- When $\Phi_B<0$
- $d\Phi_B/dt>0$
- Induced emf is nagative $\varepsilon<0$.

Step 1: Define a positive direction for the vector area A.
Step 2: From the directions of *A* and the magnetic field *B*, determine the sign of the magnetic flux ϕB and *its* rate of change.
Step 3: Determine the sign of the induced emf. If the flux is increasing, so $d\phi B/dt$ is positive, then the induced emf or current is negative; if the flux is decreasing, $d\phi B/dt$ is negative and the induced emf or current is positive.

8.1.1 direction of the induced emf



(a) A flat surface of area A bounded by a conducting loop. The positive direction of circulation around the loop is related to the vector A by the right-hand rule; (b) A steady magnetic field now pierces the loop, the magnetic flux through the loop is positive; (c) The magnitude of the magnetic field increases with time. The induced emf is in the negative direction of circulation.

8.1.2 Lenz's law: induced current direction

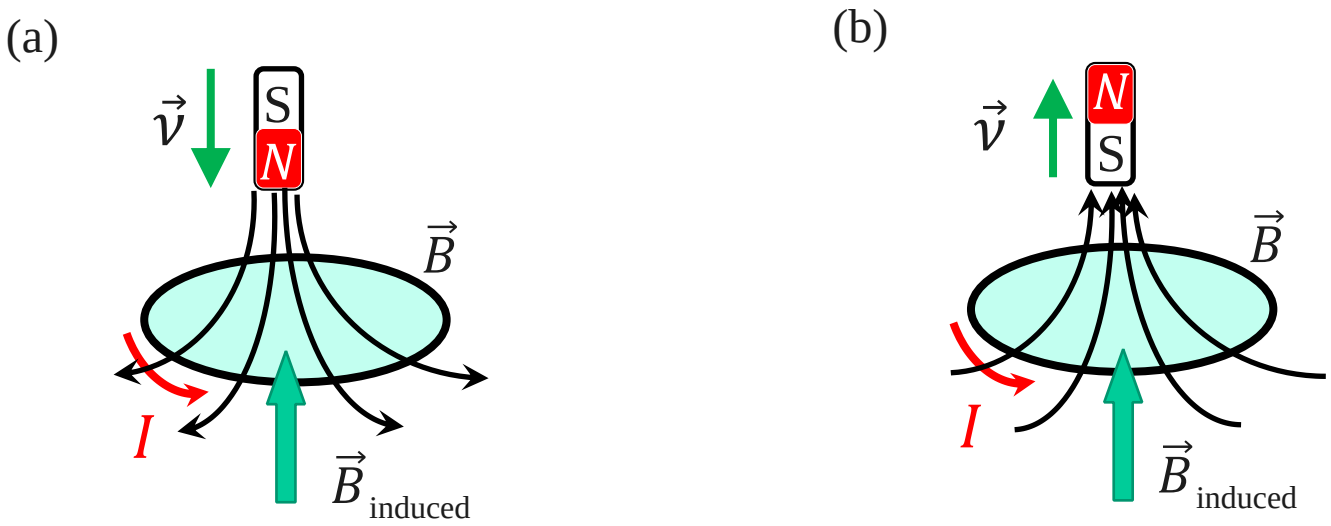
Expression 1:
The induced current in a closed conducting loop will appear in such a direction that it opposes the change that produced it.

Expression 2:
The direction of any magnetic induction effect is such as to oppose the cause of the effect.

Or in other words (not formal but easy to remember):

The total magnetic flux Do not want to change.

8.1.2 Lenz's law: induced current direction



Three steps:

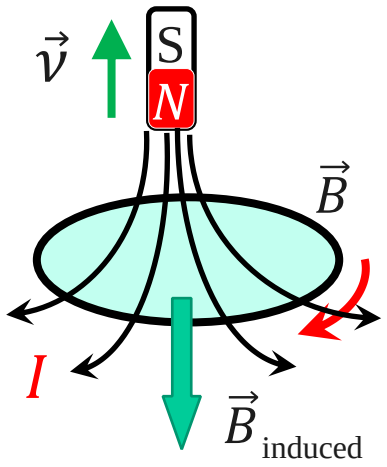
1. Judge the flux line direction through the loop. (N to S). Up or down.

2. Up is increase, then B induce is down. Up is decrease, then B induce is up. Down is increase, B induce is up, Down is decrease, then B induce is down. (because the total flux does not want to change)

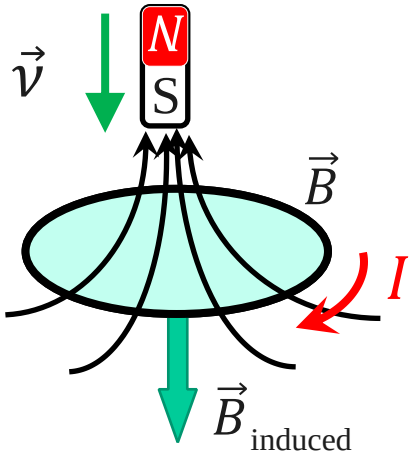
3. Then judge the I direction. (use your right hand, your thrumb finger towards to B, four fingers towards to current).

8.1.2 Lenz's law: induced current direction

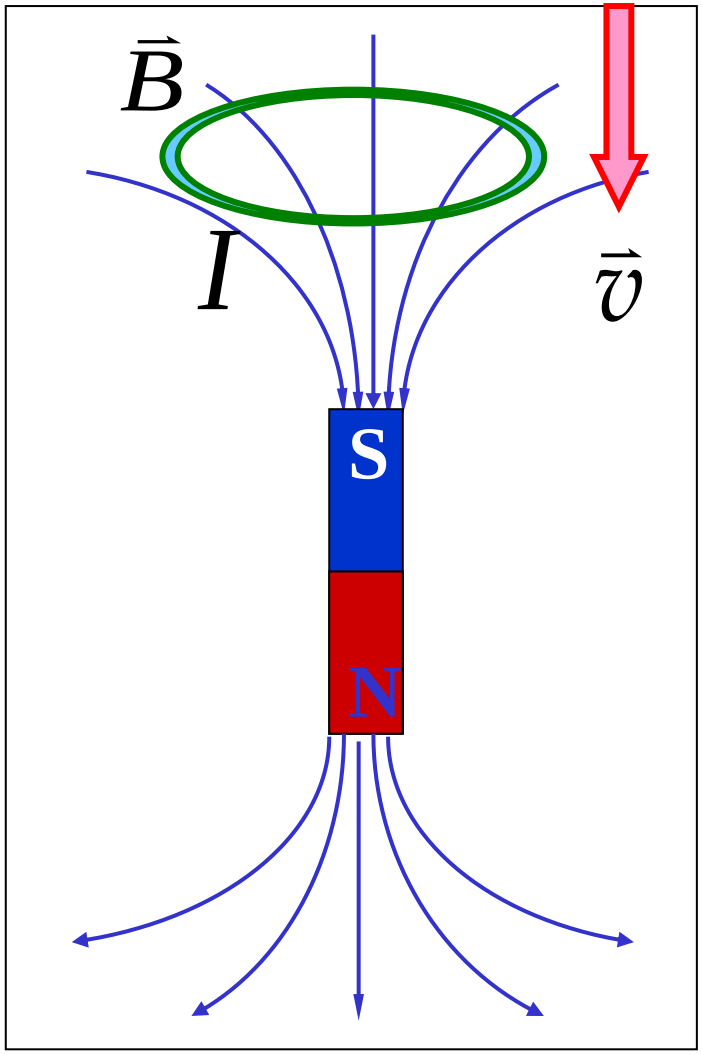
(c)

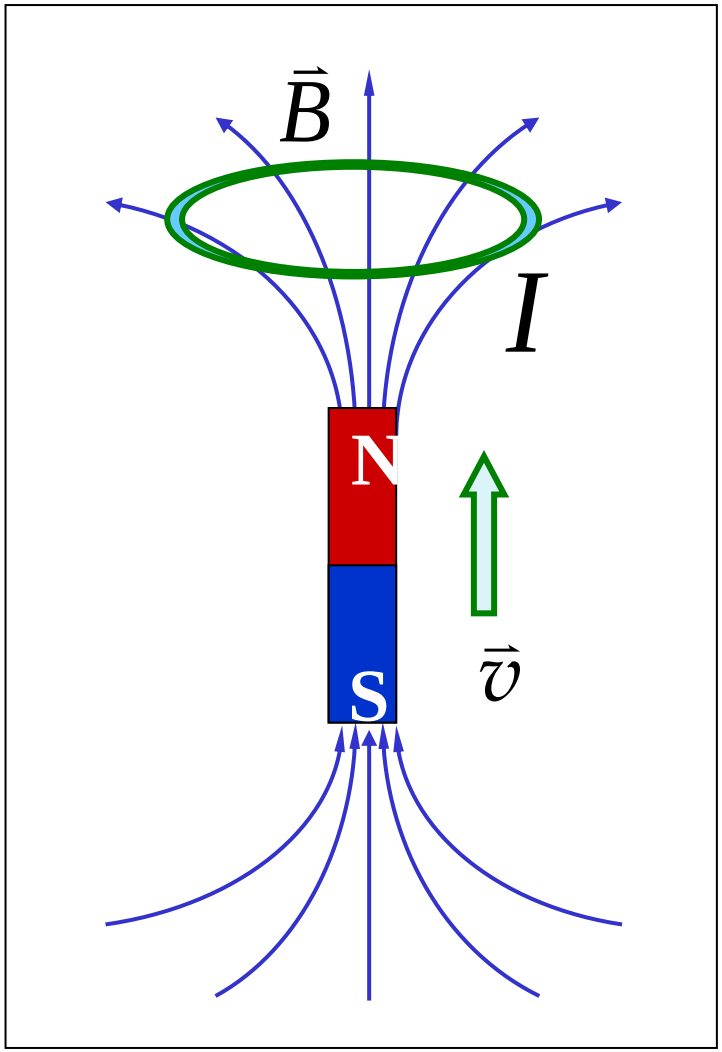


(d)



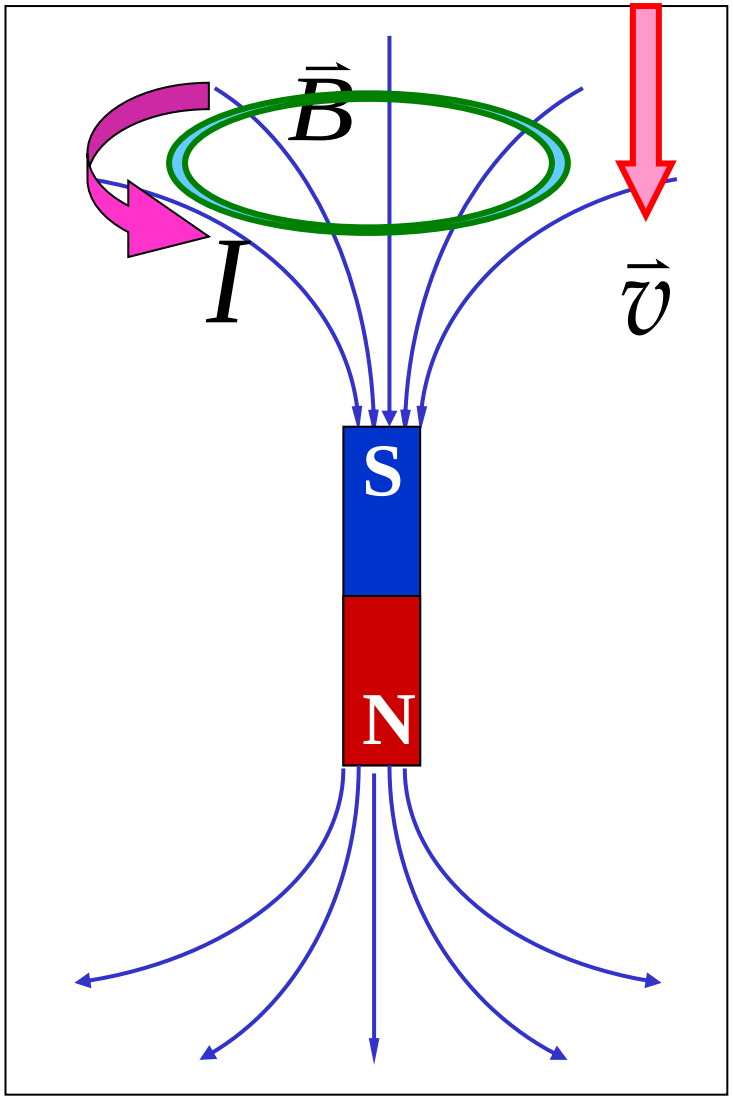
8.1.2 Lenz's law: case study

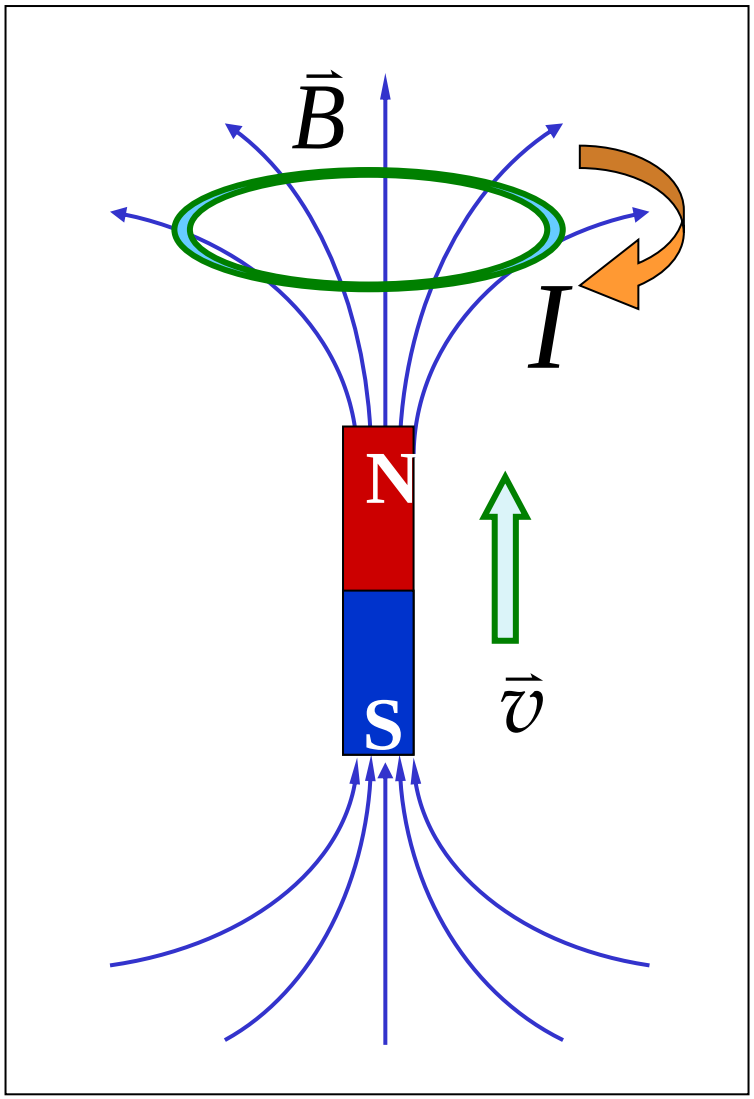




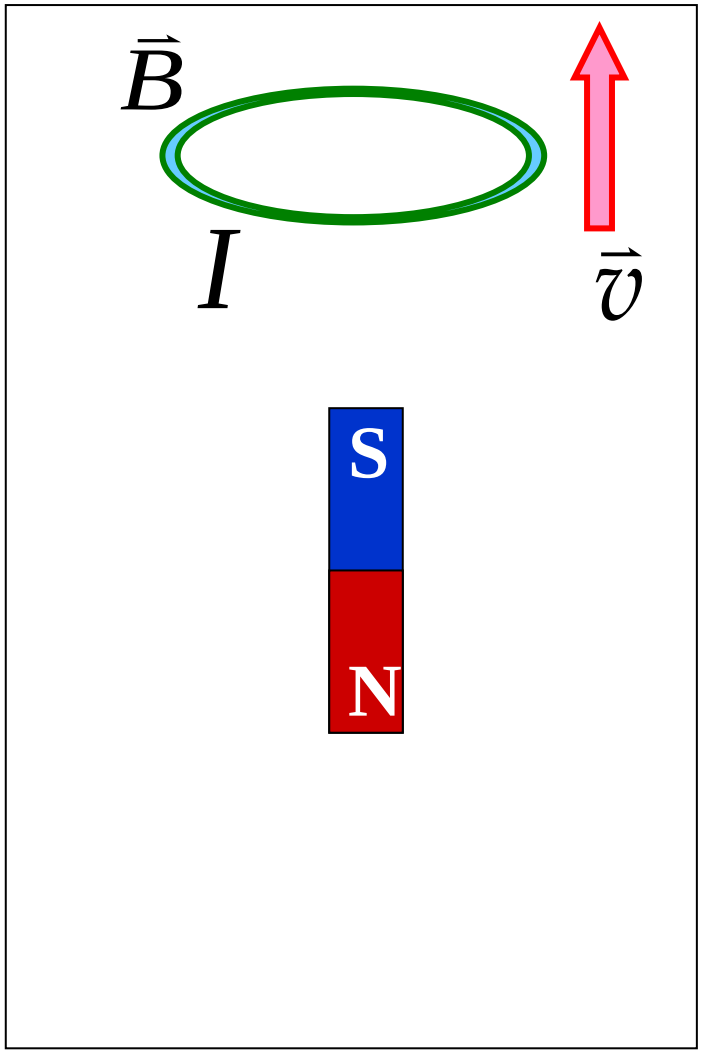
I ? direction

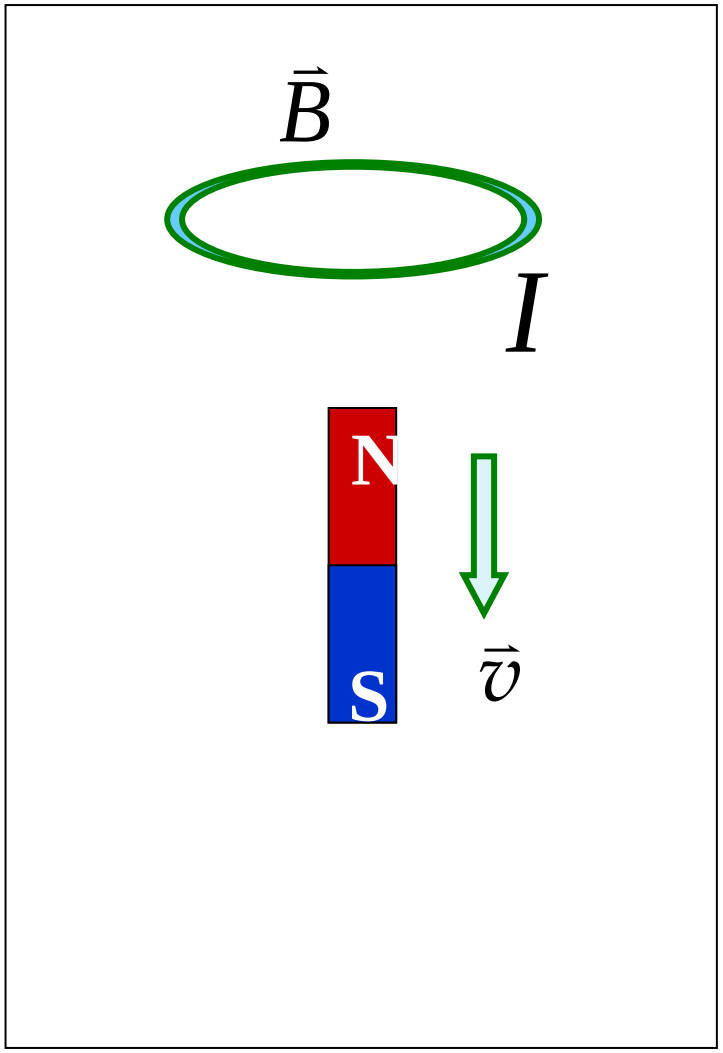
8.1.2 Lenz's law: induced current direction



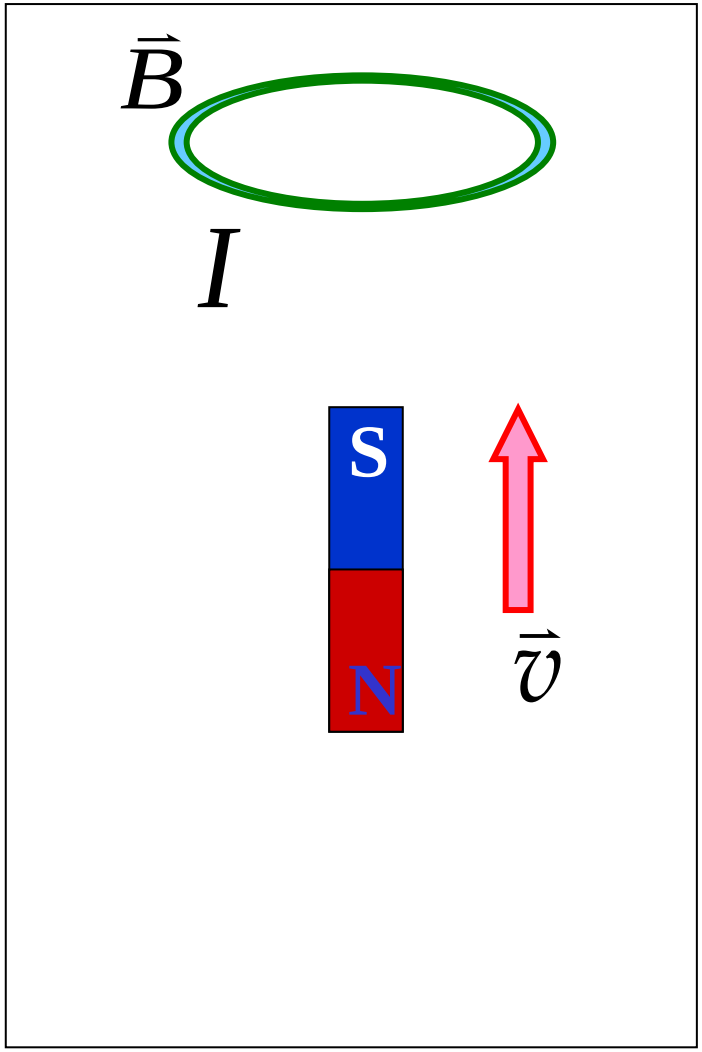


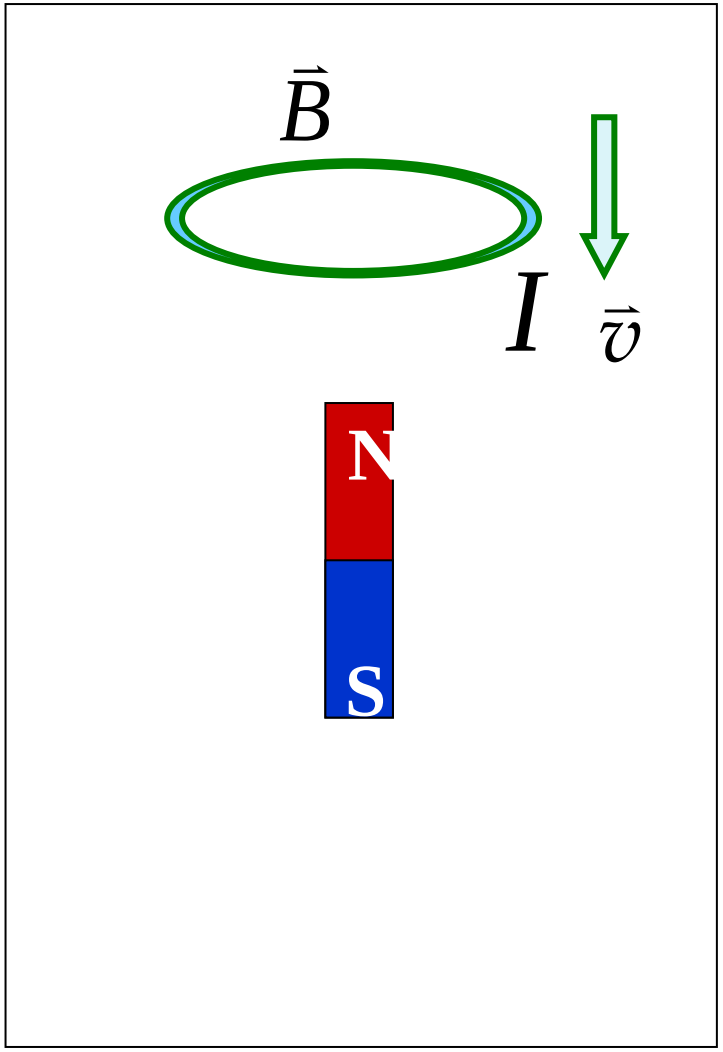
More practice



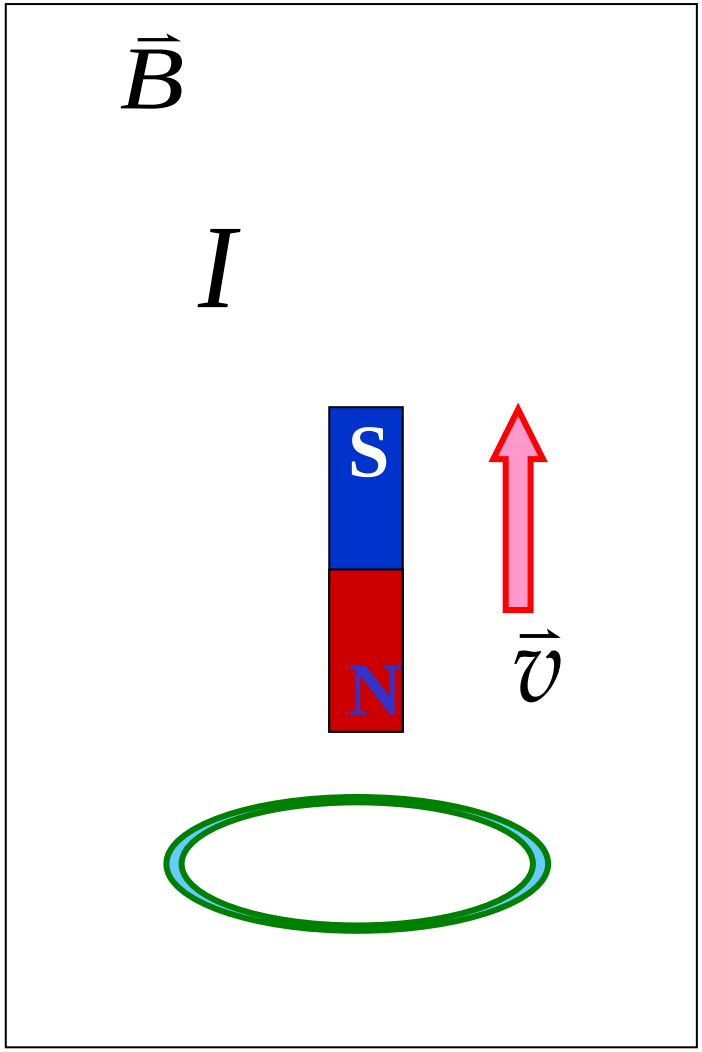


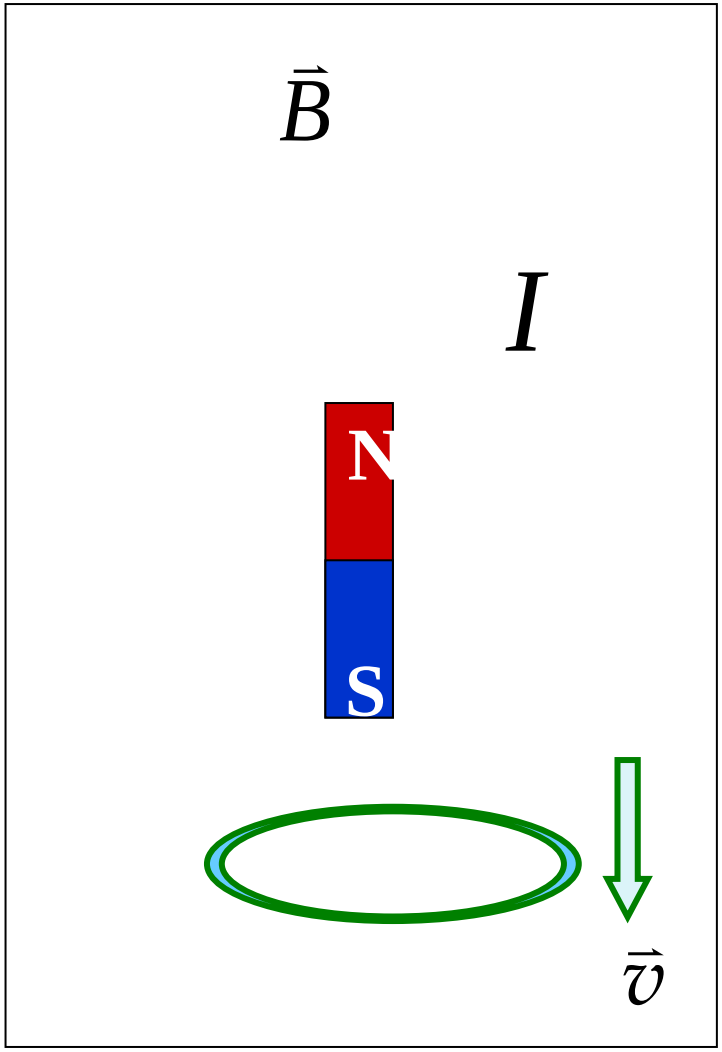
More practice





More practice





Homework

Week 5

Page 183-186 Q: 1

Homework file name:

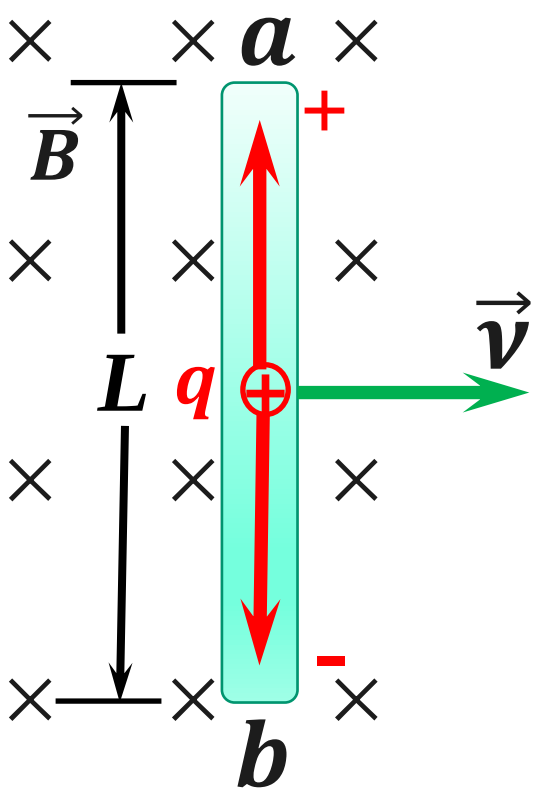
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for example: 202380090001 + week 5 + Alfa

In pdf file, send to the cloud in our group

1. A 1000-turn coil is 10cm in diameter. What emf is developed across the coil if the magnetic field through the coil is reduced from 0.6T to 0 in (a) 1s, (b) 0.1s, or (c) 0.01s.

8.1.3 Motional electromotive force



$$F_n = qv \times B$$

$$F_n = qEn$$

$$E_n = v \times B$$

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

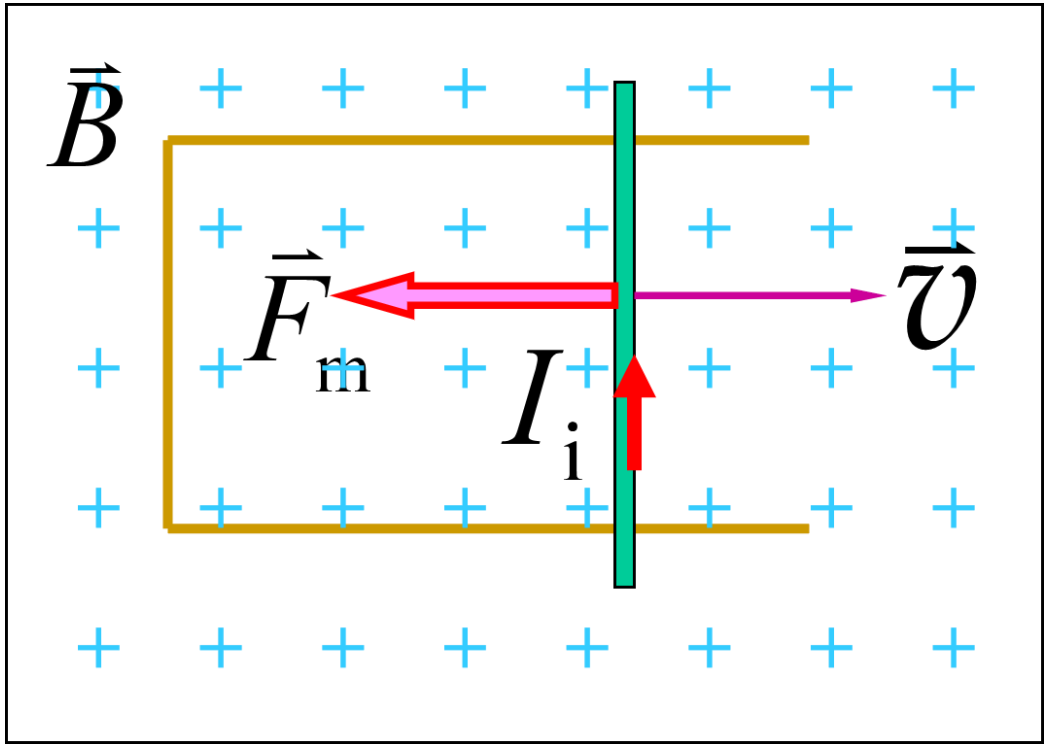
$$= \frac{BdA}{dt} = \frac{Blds}{dt} = Blv$$

8.1.3 Motional electromotive force

A general expression:

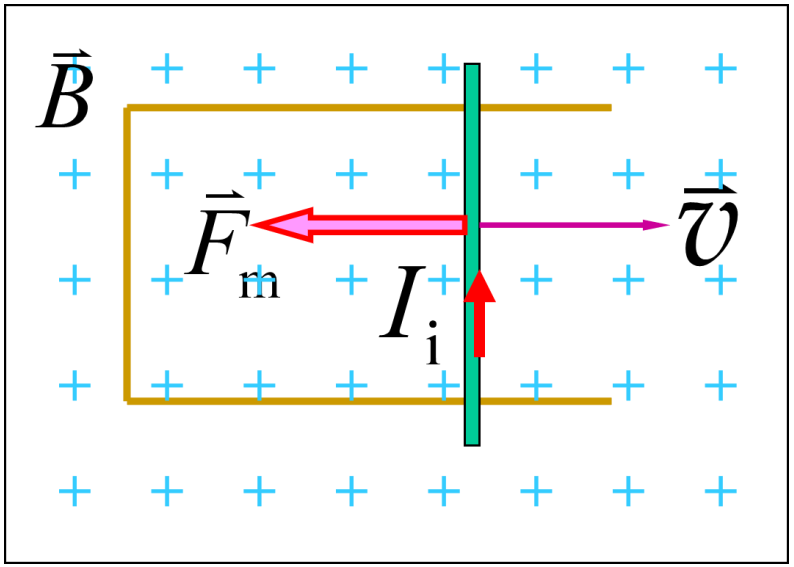
$$\varepsilon = \int_b^a v \times B \bullet dl$$

8.1.3 an example



$$\mathbf{V_{ab}} = El = vBl$$

8.1.3 Example 1



Example 1. Let the length l is 0.1m, the velocity v be 2m • s^{-1}, the resistance of the loop be 0.1Ω, and let B = 0.5T. Find (a) the motional emf (b) the current I in the loop, and, (c) the force that maintains the motion of conductor.

Solution:

(a) The motional emf is

$$\varepsilon = Blv = (0.5\ \mathrm{T}) * (0.1\ \mathrm{m}) * (2\ \mathrm{m\ s^{-1}}) = 0.1\ \mathrm{V}$$

(b) Ohm's law

$$I = \frac{\varepsilon}{R} = \frac{0.1\ V}{0.1\ \Omega} = 1\ A$$

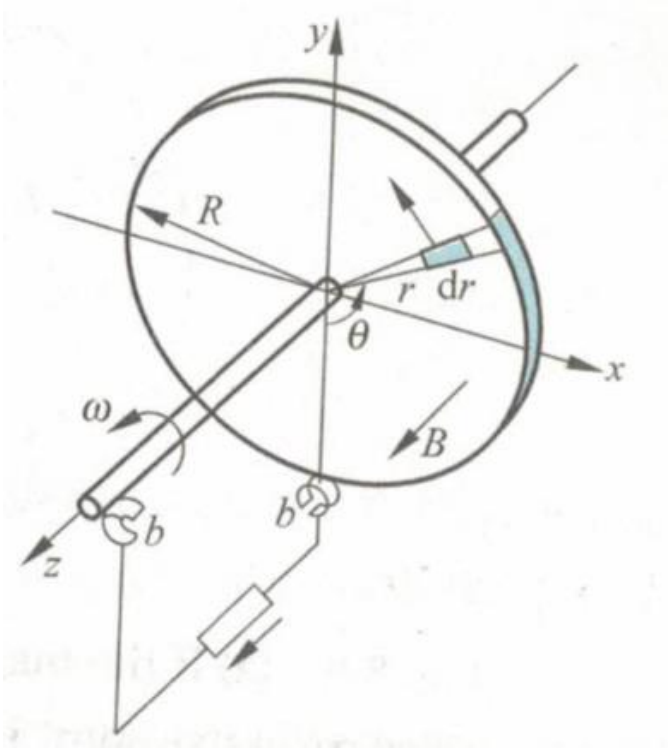
(c) The force

$$F = I \int dl \times B = IBL = (1\ A) * (0.5\ \mathrm{T}) * (0.1\ \mathrm{m}) = 0.05\ \mathrm{N}$$

8.1.3 Example 2

A disk of radius *R* lies in the xy-plane rotates with uniform angular velocity ω about the z-axis. The disk is in a uniform magnetic field *B* parallel to the z-axis

Find the emf ε between the center and rim shown in the Fig.



Faraday disk Dynamo

Solution:

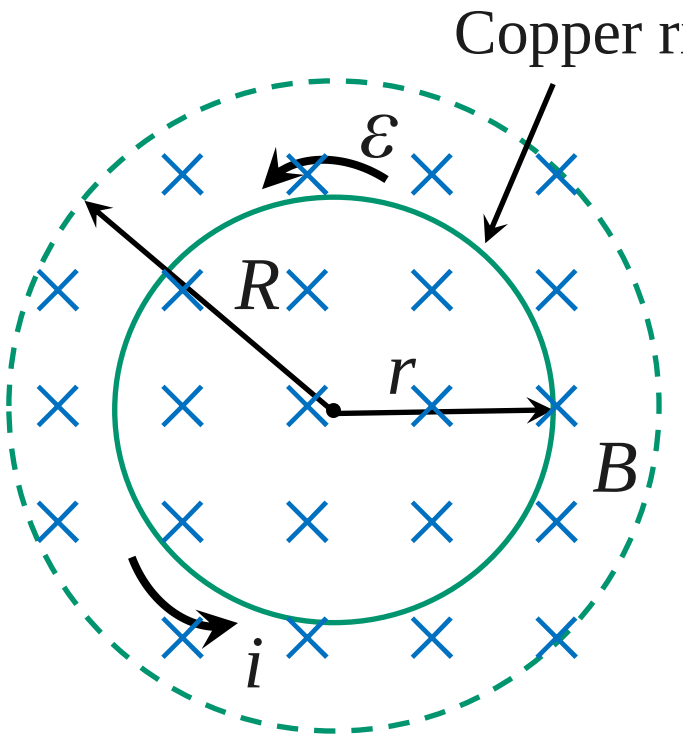
Consider a short portion of narrow radial segment of the disk, of length dr. Its velocity is $v = \omega r$

$$\varepsilon = \int_b^a v \times B \bullet dl = \int_0^R \omega r B \bullet dr$$

$$\varepsilon = \int_0^R \omega r B \bullet dr$$

$$= \omega B \int_0^R r \bullet dr = \frac{1}{2} \omega B R^2$$

8.1.4 Induced Electric Field



A changed magnetic flux, increasing with a steady rate for example

What is the direction of the current generated in the copper ring?

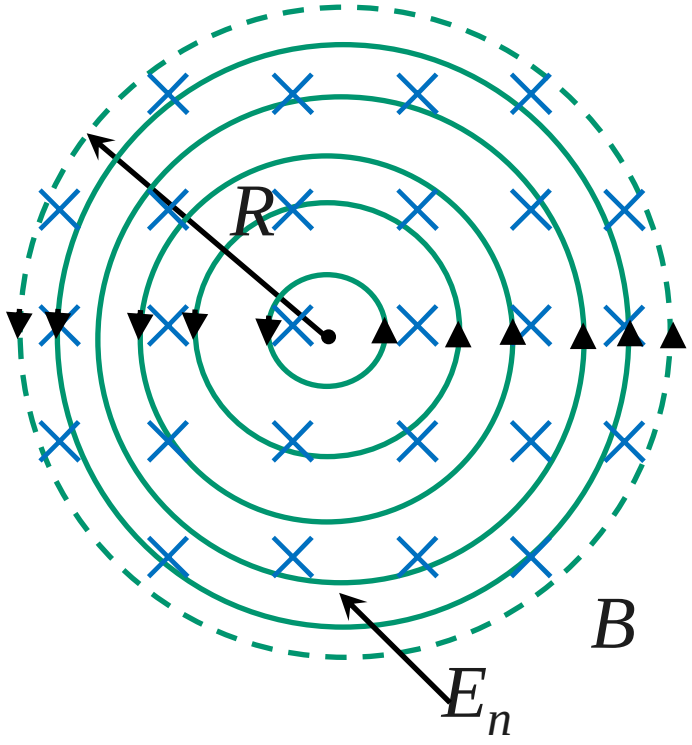
$$\varepsilon = -\frac{d\Phi_B}{dt} = \oint E_n \bullet dl$$

$$\varepsilon = \int_b^a E_n \bullet dl =$$

$$\int_b^a v \times B \bullet dl$$

A changing magnetic field produces an electric field

8.1.4 Induced Electric Field

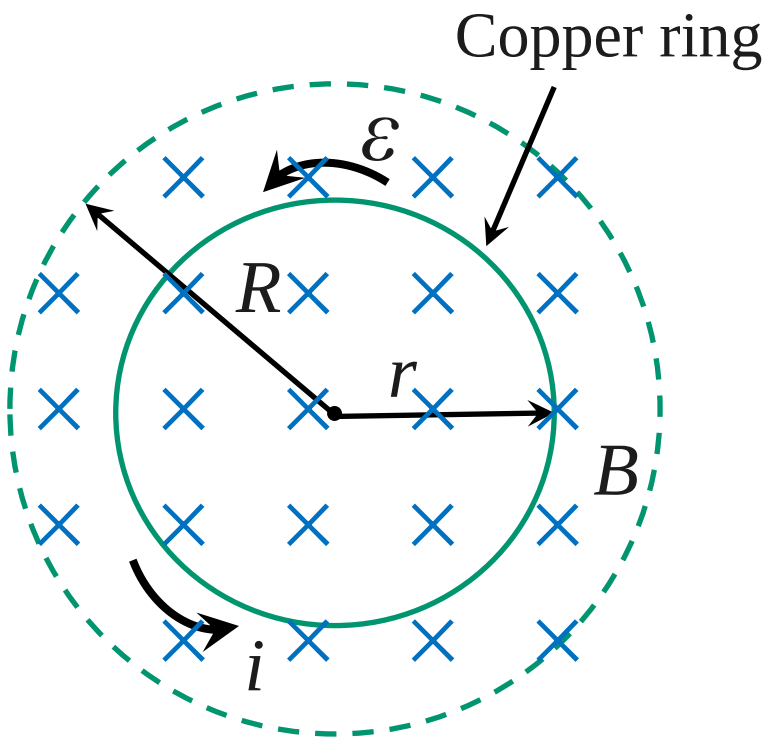


Induced electric field generated by the changed magnetic field

Even there is no copper ring!

Means: if there is a charged particle, particles will move!

8.1.4 Example

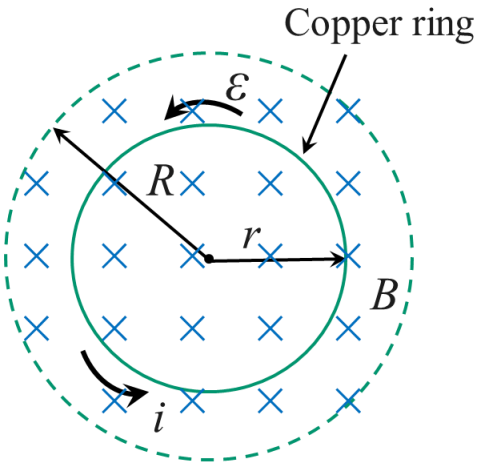


Assume that R = 8.5cm and that dB/dt = 0.13T S^{-1}

(a) What is the magnitude of the induced electric field E_n for r = 5.2cm?

(b) What is the magnitude of the induced electric field for r = 12.5cm?

Solution:



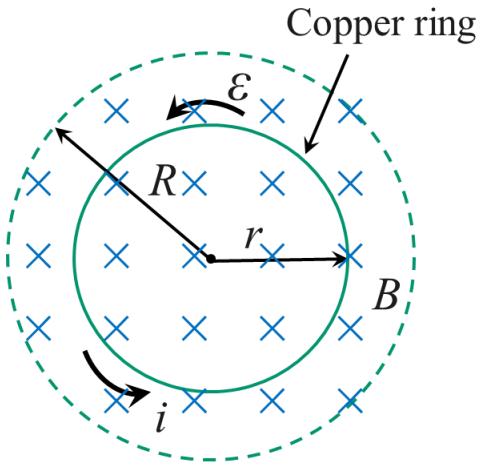
(a) $$\varepsilon = -\frac{d\Phi_B}{dt} = \oint E_n \bullet dl$$

$$-\pi r^2 \frac{dB}{dt} = E_n(2\pi r)$$

$$E_n = \frac{1}{2}\left(\frac{dB}{dt}\right)r = (0.5)\bullet(0.13\ \mathrm{T\ s^{-1}})(5.2 \times 10^{-2}\mathrm{m}) = 0.0034\ \mathrm{V\ m^{-1}}$$

(b)

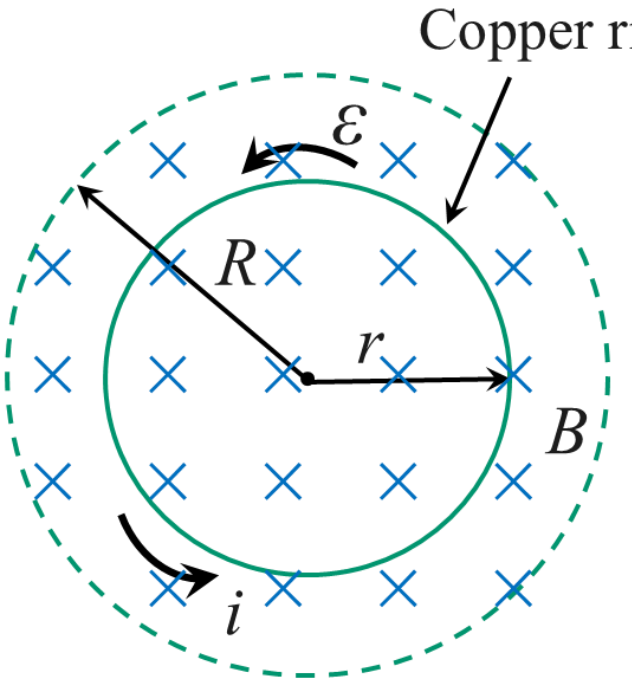
$$\varepsilon = -\frac{d\Phi_B}{dt} = \oint E_n \bullet dl$$



$$-\pi R^2 \frac{dB}{dt} = E_n(2\pi r)$$

R=8.5 cm, r=12.5 cm

$$E_n = \frac{1}{2r} R^2 \frac{dB}{dt} = 3.8 \text{ mV m}^{-1}$$

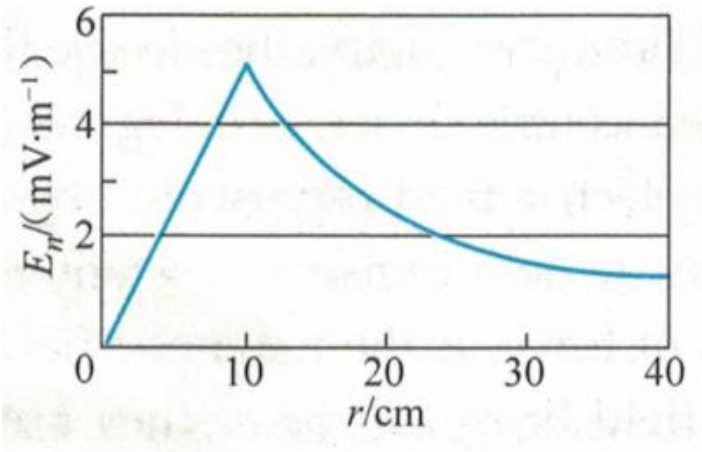


(a) r<R

$$E_n = \frac{1}{2}\left(\frac{dB}{dt}\right)\mathbf{r}$$

(b) r>=R

$$E_n = \frac{1}{2r}\left(\frac{dB}{dt}\right)R^2$$



Homework

Week 6 Page 183-186

Q: 4, 7

Homework file name:

Roll number+week 6 + name,

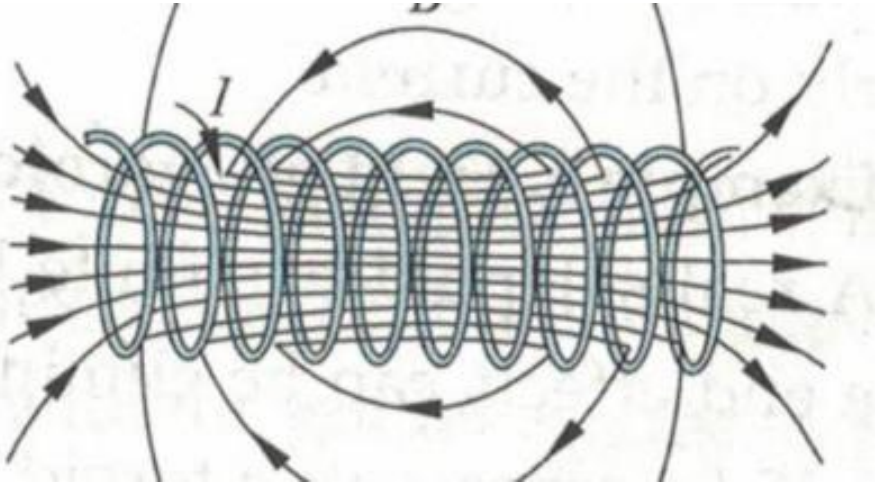
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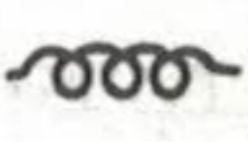
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202280090116 Md Khiruzzaman for Software Engineering),

8.2 Inductance

Inductor: can built up a magnetic field in a specified region. Such as solenoid.



Symbol 

$$L = \frac{N\phi}{I} = \frac{\psi}{I}$$

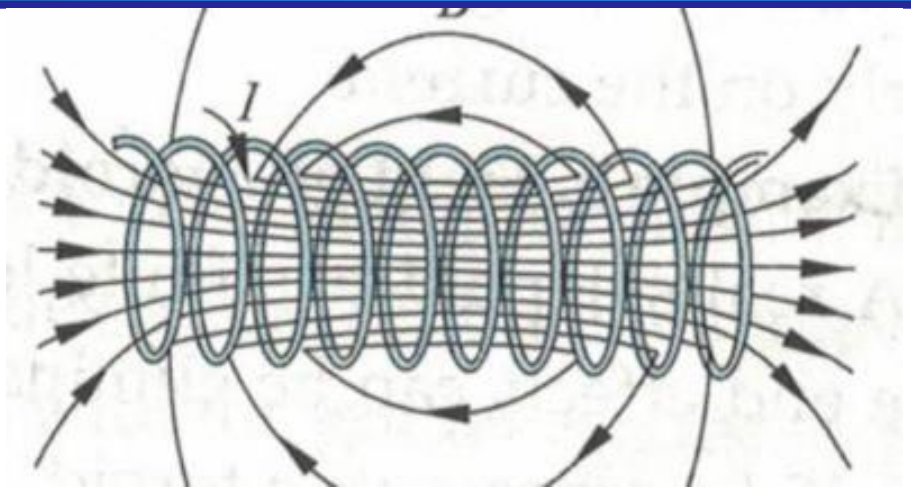
N: number of turns

$N\phi = \psi$: number of flux linkage

L: unit: henry : H

ϕ: $unit$: $weber = tesla\ meter^2$

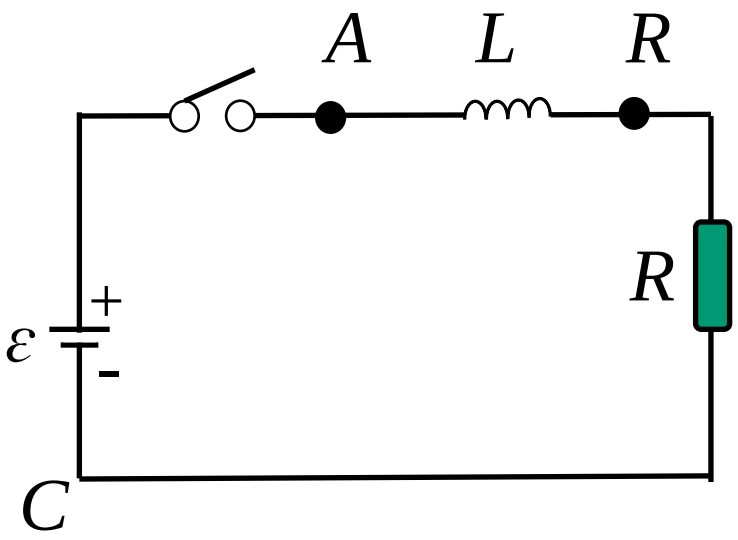
8.2 Inductance



$$\varepsilon = -\frac{d\psi}{dt} = -N\frac{d\phi}{dt} = -L\frac{dI}{dt}$$

$$L = \frac{N\phi}{I} = \frac{(nl)BA}{I} = \frac{(nl)\mu_0 nIA}{I} = \mu_0 n^2 lA$$

8.2.2 Magnetic Energy and the magnetic Field



$$\varepsilon = IR + L\frac{dI}{dt}$$

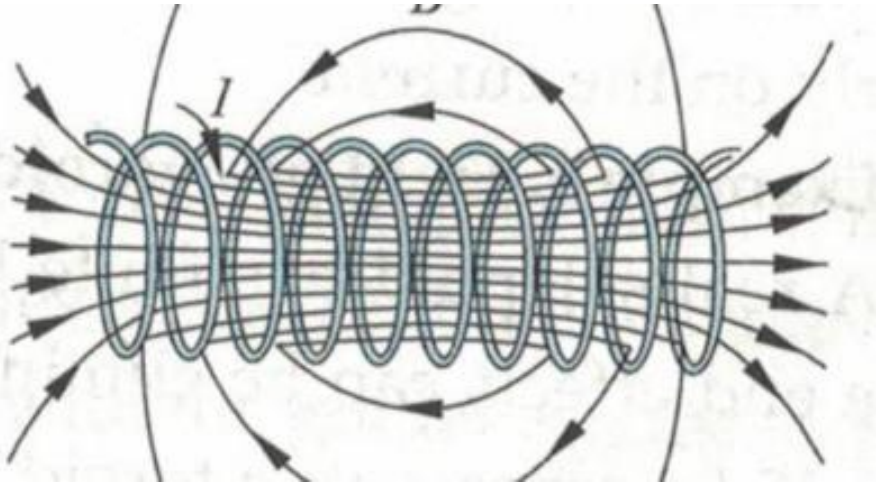
$$\varepsilon I = I^2R + LI\frac{dI}{dt}$$

$$\frac{dU}{dt} = LI\frac{dI}{dt} \qquad dU = LIdI$$

$$U = \int \mathbf{d}U = \int_0^l LIdI$$

$$U = \frac{1}{2}LI^2$$

8.2.2 Magnetic Energy and the magnetic Field



$$B = \mu_0 nI$$

$$L = \mu_0 n^2 l A$$

$$U = \frac{1}{2} LI^2 = \frac{1}{2} (\mu_0 n^2 l A)\left(\frac{B}{\mu_0 n}\right)^2 = \left(\frac{1}{2\mu_0}\right)(Al)B^2$$

Magnetic energy density

$$u = \frac{U}{Al} = \frac{B^2}{2\mu_0}$$

$$U = \int \mathbf{d}U = \int u dV = \int \frac{B^2}{2\mu_0} dV$$

8.2.2 Example

Example 1. Compare the energy required to set up, in a cube 10 cm on edge (a) a uniform electric field of 100 kV • m^{-1} and (b) a uniform magnetic field of 1.0 T. Both these fields would be judged reasonably large but readily available in the laboratory.

8.2 Inductance

Solution:

(a) In the electric case we have, where V_0 is the volume of the cube,

$$U_E = u_E V_0 = \frac{1}{2}\varepsilon_0 E^2 V_0$$

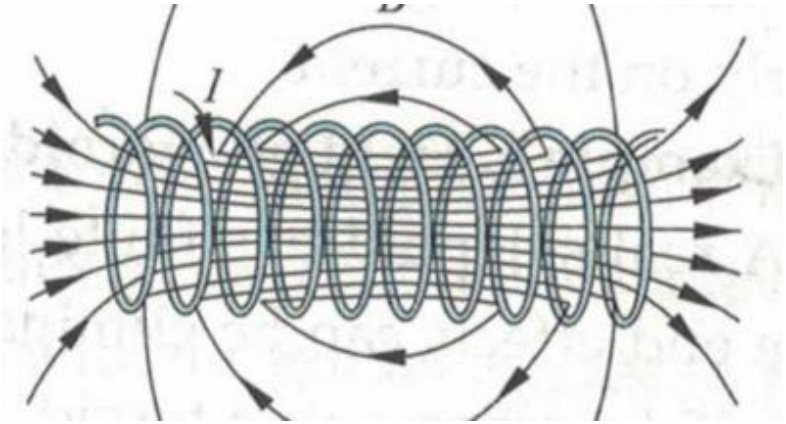
$$= \left(\frac{1}{2}\right)(8.85 \times 10^{-12}\ \text{F m}^{-1})(10^5\ \text{V m}^{-1})^2(0.1\ \text{m})^3 = 4.4 \times 10^{-5}\ \text{J}$$

(b) In the magnetic case, we have

$$U_B = \mu_B V_0 = \frac{B^2}{2\mu_0} V_0$$

$$= \frac{(1.0\text{T})^2(0.1\ \text{m})3}{2(4\pi \times 10^{-7}\ \text{T m A}^{-1})} = 398\ \text{J}$$

8.3 Magnetic Properties of Matter



$$B_E = \mu_0 nI = \mu_0 \left(\frac{N}{l}\right) I$$

The magnetic dipole moment

$$\mu = NIA$$

The torque

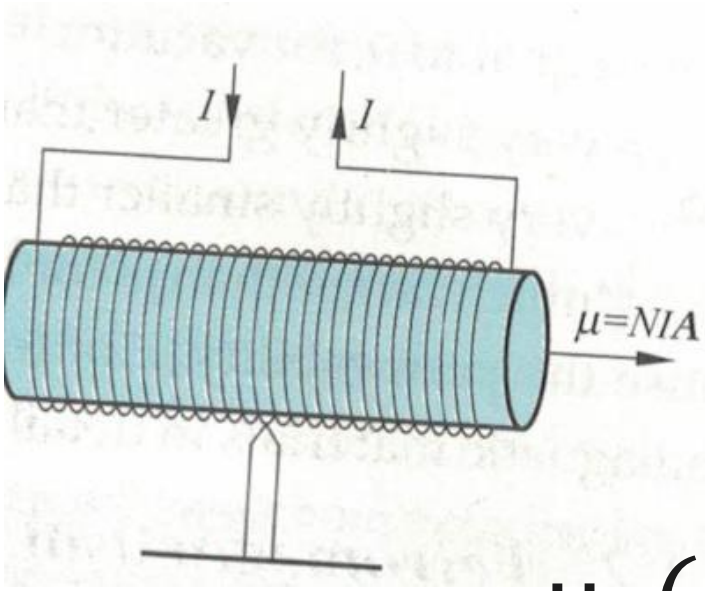
$$\boldsymbol{\tau} = \mu \times B$$

$$B_E = \mu_0 nI = \mu_0 \left(\frac{NIA}{Al}\right) = \frac{\mu_0 \mu}{V}$$

The Volume

$$V = Al$$

8.3.1 Classification of magnetic materials



When insert some materials, the induced magnetic moments produce an additional contribution to the magnetic moment

$$B = \frac{\mu_0(\mu + \mu_l)}{V} = \frac{\mu_0 K_m \mu}{V}$$

$$K_m = \frac{B}{B_E}$$

Relative permeability

Permeability of the materials

(a) equal to μ_0 : vacuum

(b) paramagnetism: very slightly greater than μ_0

(c) diamagnetism : very slightly smaller than μ_0

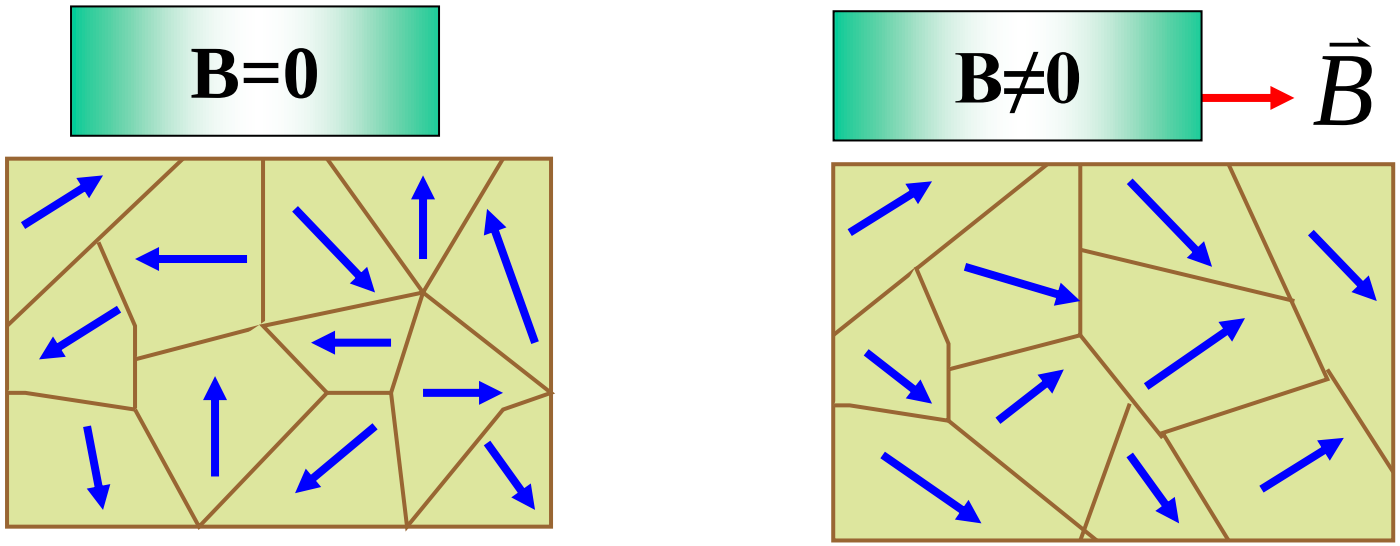
(d) ferromagnetism: much larger than μ_0

8.3.2 ferromagnetism: $\mu' \gg \mu_0$

Ferromagnetic Materials, 5 elements and with their alloy

Iron
Cobalt
Nickel
Dysprosium
Gadolinium

8.3.2 ferromagnetism: $\mu' \gg \mu_0$



There are regions in every ferromagnetic sample has near perfect alignment of magnetic dipole moment even when there is no applied magnetic field.

When applied a magnetic field, some domains are grown bigger and bigger

Domain size: 10^{-18} to 10^{-12} m^3

8.3.2 ferromagnetism: $\mu' \gg \mu_0$

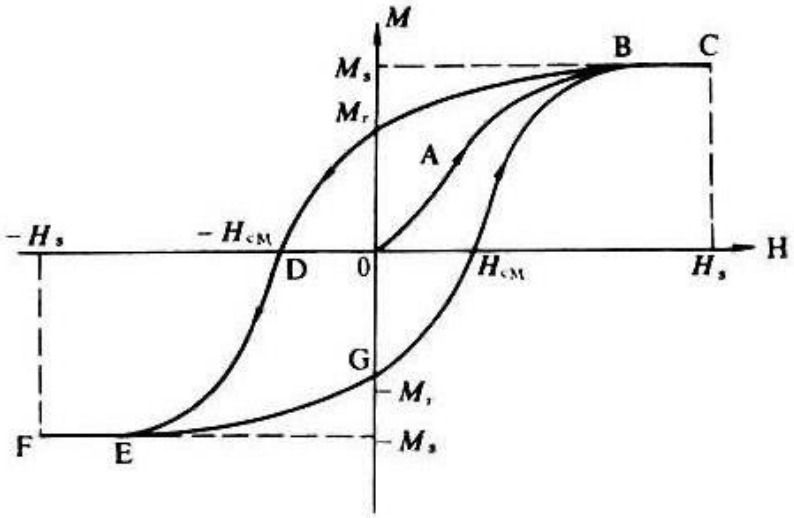
Curie temperatures of the ferromagnetic elements

Element	T_c(K)
Iron	1043
Cobalt	1404
Nickel	631
Gadolinium	289
Dysprosium	85

When T > curie temperature
The material is only paramagnetic.

8.3.2 Hysteresis loop of ferromagnetic samples

Hysteresis: memory



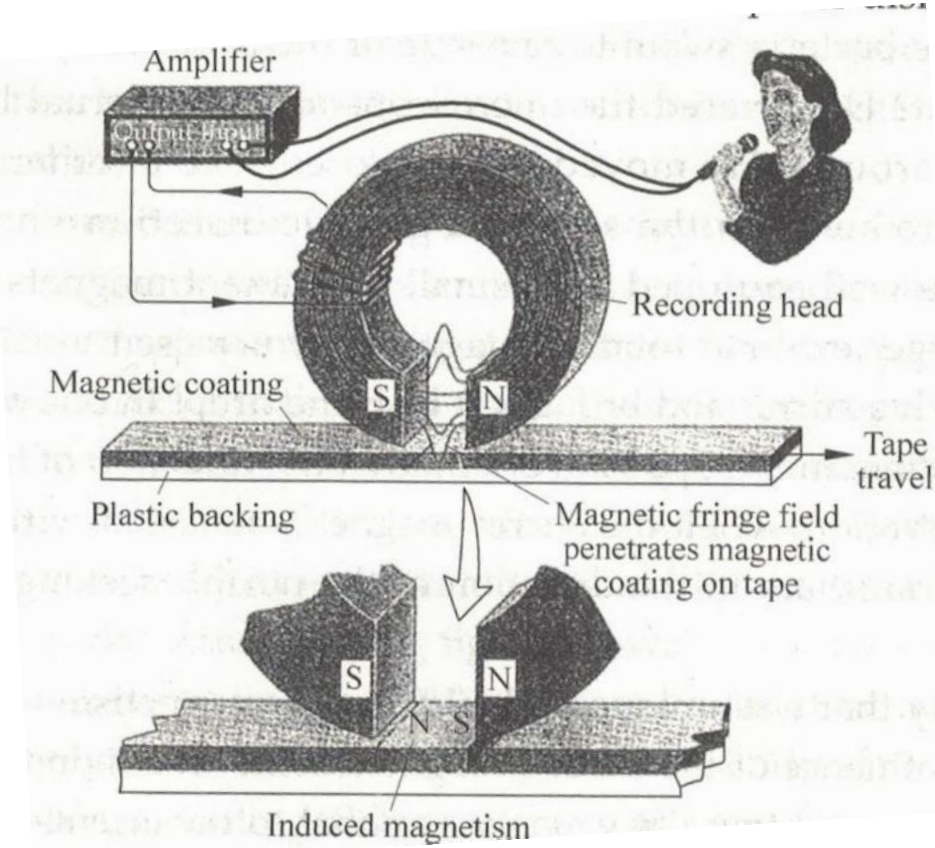
Hysteresis can be understood on the basis of the magnetic domain concept. When the applied current *I* (and therefore the magnetic field *BE)* is increased and then decreased back to its initial value the domains do not return completely to their original configuration but retain some “memory” of the initial increase.

Application: Tape, Ferromagnetic



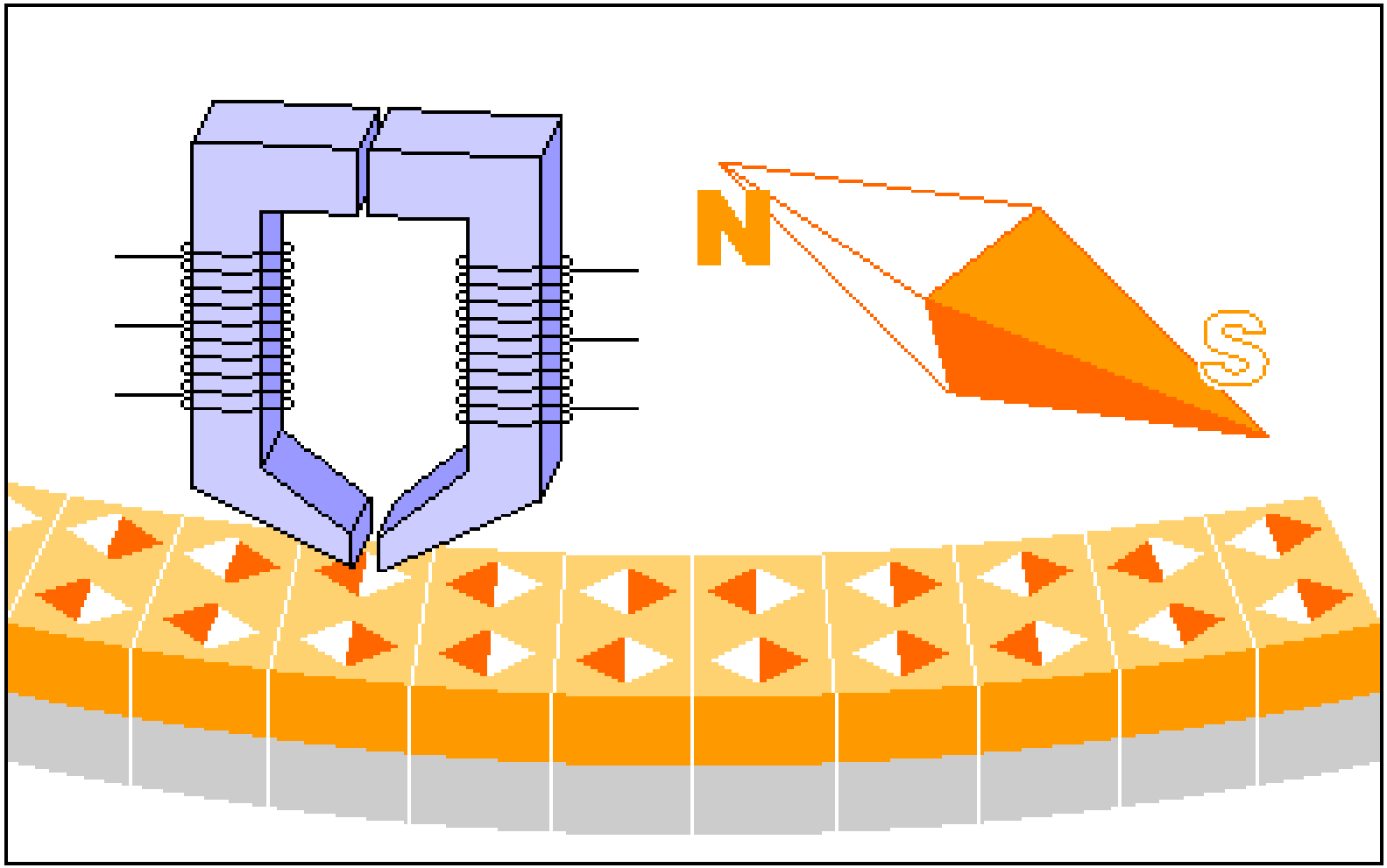


8.3.2 Working principle of tape



The “memory” of magnetic materials is essential for the magnetic storage of information, as on cassette tapes or computer disks.

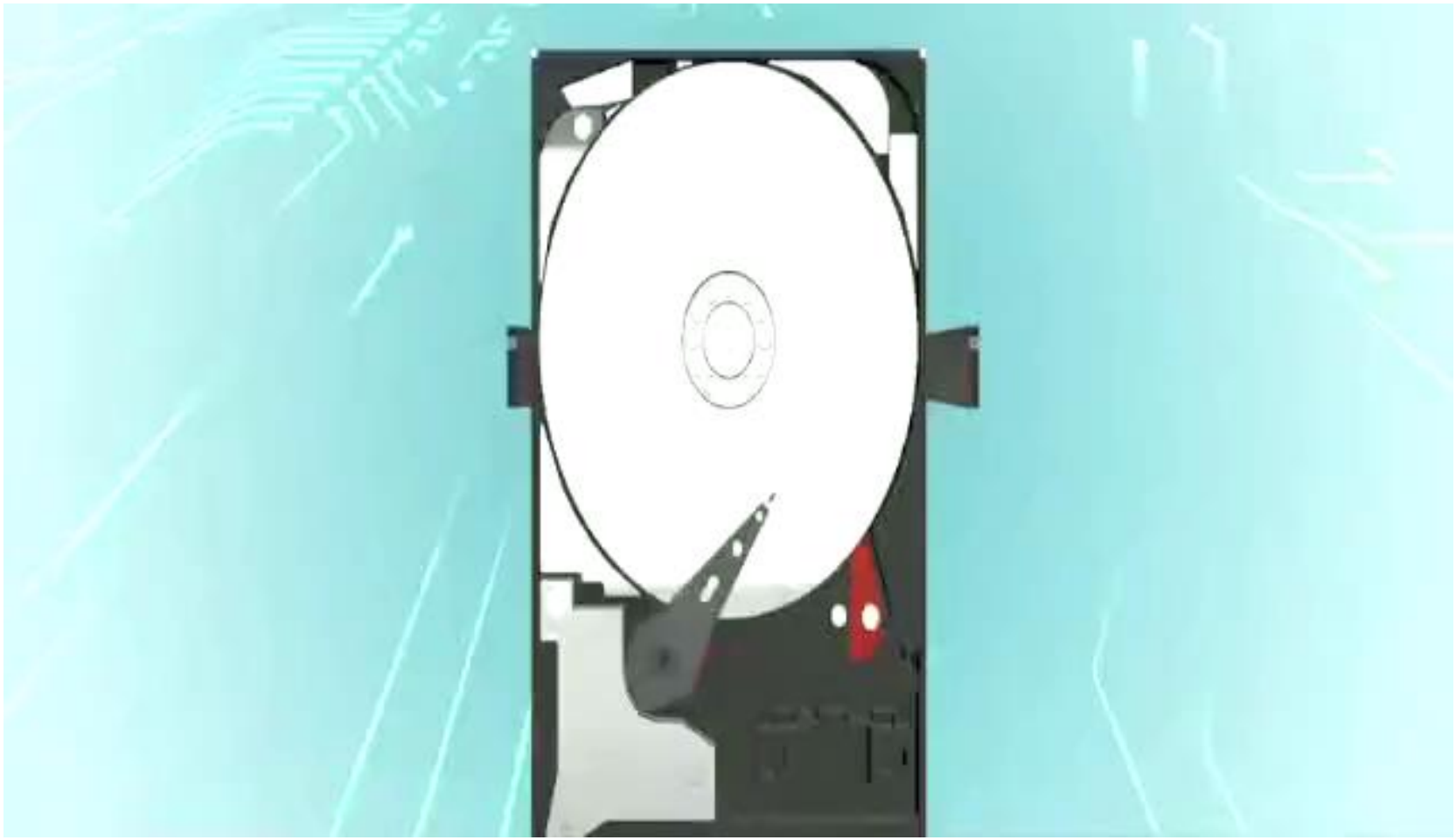
Application: GMR, Giant Magnetoresistance



Application: hard disk



Application: hard disk



Homework

Week 7 Page 183-186

Q: 9, 10, 12, 13, 14, 17, 20

Homework file name:

Roll number+week 7 + name,

For example 202280090199 + week 3 + Alfa Robot

In pdf file, send to monitors (Usman – 202280040103 for Civil Engineering 202280090116 Md Khiruzzaman for Software Engineering),

8.4 Electromagnetic Waves

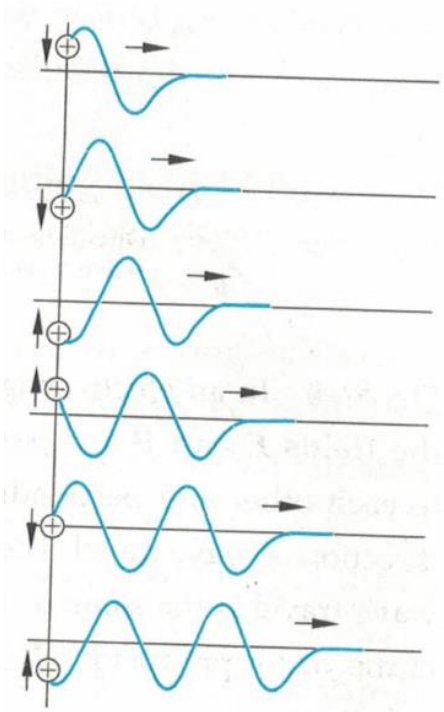


Fig.8-17 As the charge oscillates up and down, changes in the *E*-field travel outward as a wave.

The two planes containing the E-field and B-field of an electromagnetic wave are mutually perpendicular

The E-field and the B-field travel through space together at a finite speed of C (the speed of light).

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

$$C = 2.998 \times 10^8 \text{ m s}^{-1}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

8.4 Electromagnetic Waves

$$C = \frac{\lambda}{T} = \lambda \mathbf{v}$$

$$E_x = E_0 \sin[\frac{2\pi}{\lambda}(z - ct)]$$

$$B_x = B_0 \sin[\frac{2\pi}{\lambda}(z - ct)]$$

λ: *wave length*

v: frequency

T: the period

E_0 and B_0: the amplitudes of the wave

8.4 Electromagnetic Waves

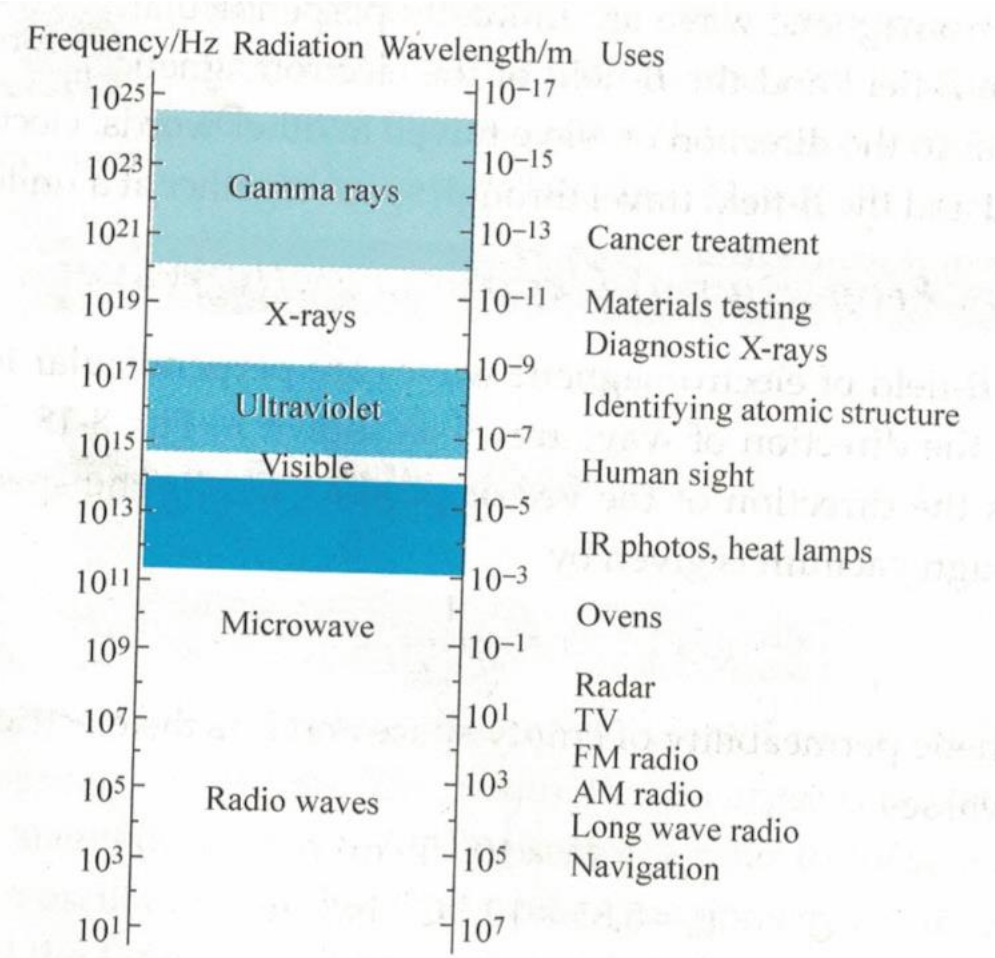
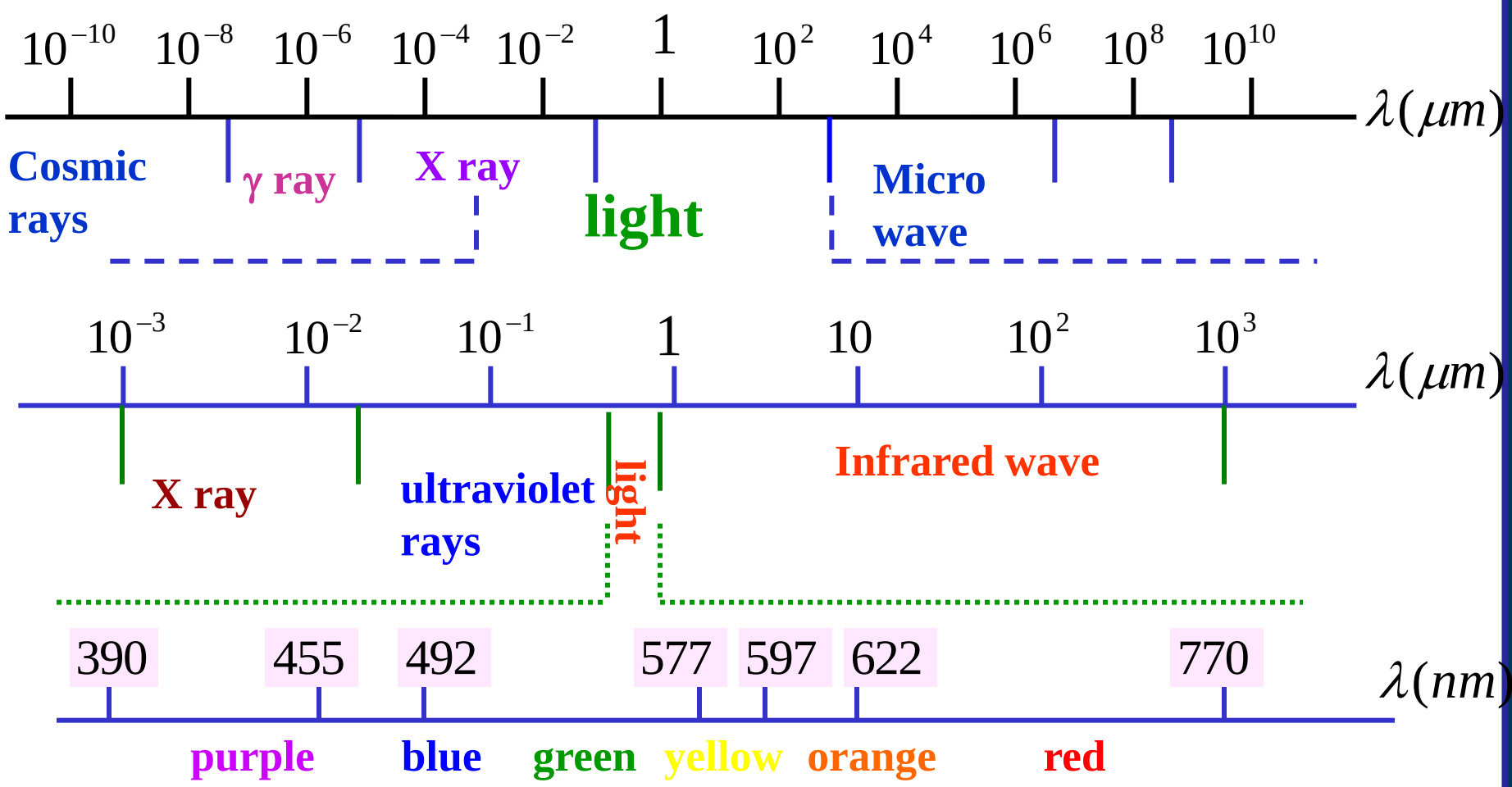


Fig. 8-19 The spectrum of electromagnetic radiation.

8.4 Electromagnetic spectrum



Questions

1. In most parts of the northern hemisphere the earth's magnetic field has a vertical component directed into the earth. An airplane flying east generates an emf between its wingtips. Which wingtip acquires an excess of electrons and which a deficiency?

Questions

2. A sheet of copper is placed between the poles of an electromagnet, so the magnetic field is perpendicular to the sheet. When it is pulled out, a considerable force is required, and the force required increases with speed. What's happening?
3. Can the magnetic field change over some region without a change in the magnetic flux through a surface in the region? If so, give as many examples as you can.

Questions

4. A changing magnetic field induces an emf in a metal ring. How would the induced emf change if the resistance of the ring were doubled? Assume that the induced current is so small that its magnetic effects can be neglected.
5. A short metal rod moves through a uniform magnetic field, cutting the field lines. Is an emf induced in the rod? Is there a magnetic force opposing the motion of the rod? Does work have to be done to maintain the motion of the rod?
6. A bar magnet falls under the influence of gravity along the axis of a long cylindrical copper tube. If air resistance and any friction against the bar magnet are neglected, will there be any forces opposing the fall of the magnet? Does the falling bar magnet reach a constant terminal velocity? Explain.

Questions

7.Can a charged particle at rest be set in motion by the action of a magnetic field? If not, why not? If so, how?

8. A short solenoid carrying a steady current is moving toward a conducting loop as in Fig. 8-20. What is the direction of the induced current in the loop as one sights toward it as shown?

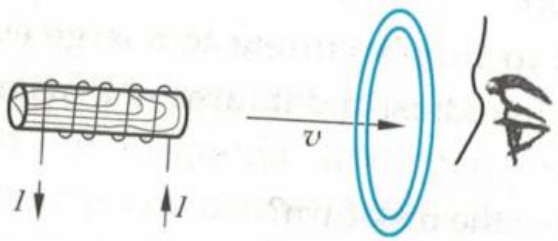


Fig. 8-20 Question 8.

Questions

9. The resistance *R* in the left-hand circuit of Fig. 8-21 is being increased at a steady rate. What is the direction of the induced current in the right-hand circuit?

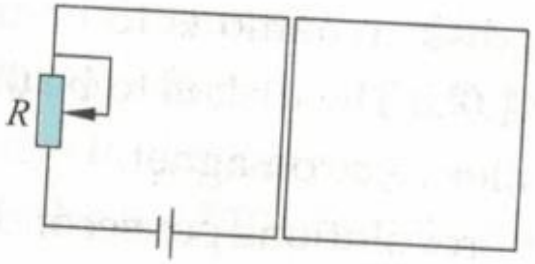


Fig. 8-21 Question 9.

10. Two closely wound circular coils have the same number of turns, but one have twice the radius of the other. How are the self-inductances of the two coil related?

Questions

11. Suppose there is a steady current in an inductor. If one attempts to reduce the current to zero instantaneously by opening a switch, a big fat arc appears at the switch contacts. Why? What happens to the induced emf in this situation? Is it physically possible to stop the current instantaneously?

12. Two iron bars are identical in appearance. One is magnet and one is not. How can you tell them apart? You are not permitted to suspend either bar as a compass needle or to use any other apparatus.

Questions

13. You want to wind a coil so that it has resistance but essentially no inductance. How would you do it?

14. When a piece of magnetic material is attracted to a permanent magnet, it is moving and hence has kinetic energy just before it strikes the magnet and sticks to it. Where does the energy come from? Where does it go?

Homework

Week 8: Page 183-186 Q17, 20

Homework file name:

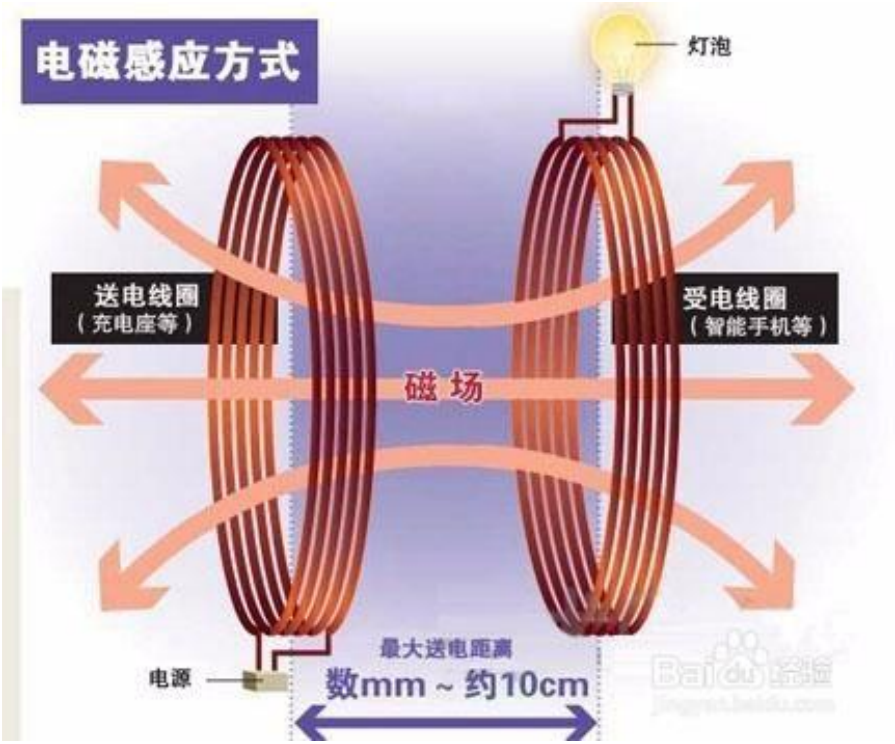
Roll number+week 8 + name,

For example 202280090199 + week 3 + Alfa Robot

In pdf file, send to monitors (Usman – 202280040103 for Civil Engineering 202280090116 Md Khiruzzaman for Software Engineering),

Wireless charge technology

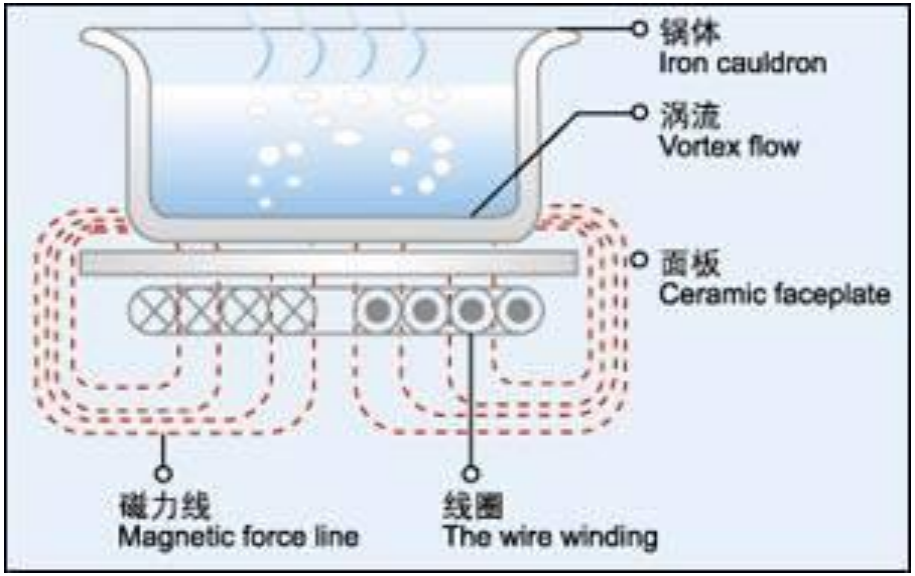




Applications

Induction cooker





Problems

1. A 1000-turn coil is 10cm in diameter. What emf is developed across the coil if the magnetic field through the coil is reduced from 0.6T to 0 in (a) 1s, (b) 0.1s, or (c) 0.01s.

the current through R?

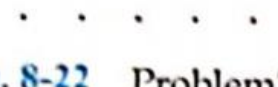


Fig. 8-22 Problem3.

4. A circular loop of wire 10 cm in radius is in a uniform magnetic field, the plane of the circular loop perpendicular to the direction of the field. The magnetic field varies with time $B = B_0 \cos\omega t$ (B_0 and ω are constant), (a) Calculate the magnetic flux through the loop and the time rate of change of this flux, (b) Calculate the emf e induced in the loop, (c) If the resistance of the loop is R , what current exists in the loop? (d) At what rate is electrical energy being dissipated in the ring?

dissipated as heat in the electromagnet?

7. Fig. 8-24 shows a uniform magnetic field B confined to a cylindrical volume of radius R. B is decreasing in magnitude at a constant rate of $10\text{mT}\cdot\text{s}^{-1}$. What is the instantaneous acceleration (direction and magnitude) experienced by an electron placed at a, at b, and at c? (Assume $r = 5.0\text{cm}$.)

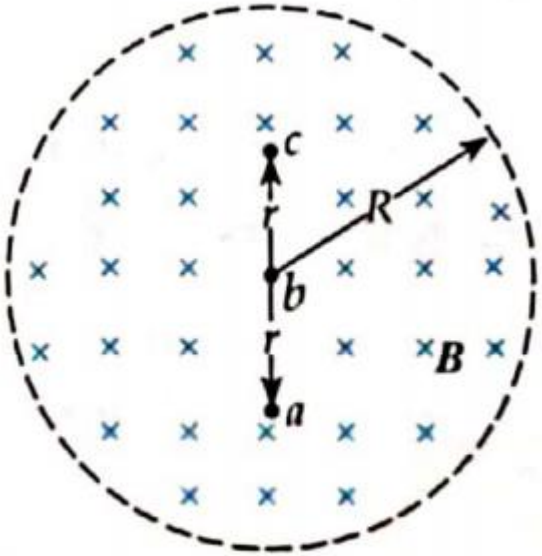


Fig. 8-24 Problem 7.

9. The current in an inductor with $L = 3\text{mH}$ is given by

$$I = 2.2\cos(60t)\text{A}$$

where t is in seconds. Determine the maximum emf induced across the inductor.

10. A long cylindrical solenoid with 100 turns per cm has a radius of 1.6cm.Assume the magnetic field it produces is parallel to its axis and is uniform in its interior. (a) What is its inductance per meter of length? (b) If the current changes at the rate $13 A \cdot s^{-1}$, what emf is induced per meter?

12. The current in the wire AB of Fig. 8-25 is upward and is increasing steadily at a di/dt.

(a) At an instant when the current is i, what are the magnitude and direction of the field B at a distance r from the wire?

(b) What is the flux $d\Phi$ through the narrow shaded strip?

(c) What is the total flux through the loop?

(d) What is the induced emf in the loop?

(e) Evaluate the numerical value of the induced emf if $a = 10\text{cm}$, $b = 30\text{cm}$, $l = 20\text{cm}$, and $di/dt = 2\text{A} \cdot \text{s}^{-1}$.

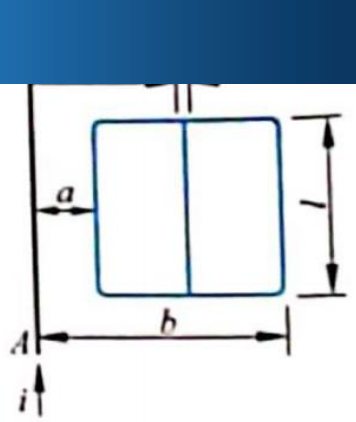


Fig. 8-25 Problem 12.

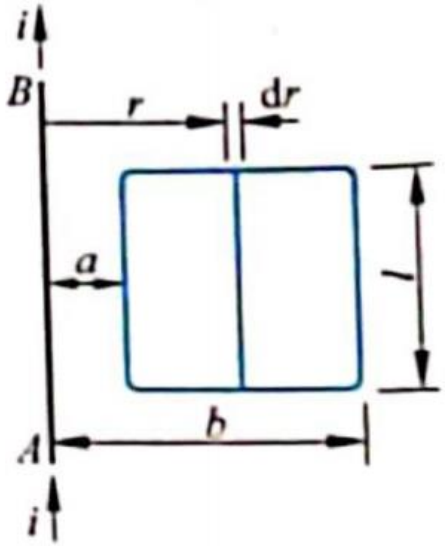


Fig. 8-25 Problem 12.

13. A straight wire carries a current $I = 10\text{A}$ near a rod that moves across two conducting wires (Fig. 8-26). The resistor has $R = 5.0\Omega$ and the rod moves at speed $3.0\text{m} \cdot \text{s}^{-1}$.

(a) What is the emf induced in the rod?

(b) What is the current in the circuit?

(c) How much work is done to move the rod 10cm to the right? What force does this work?

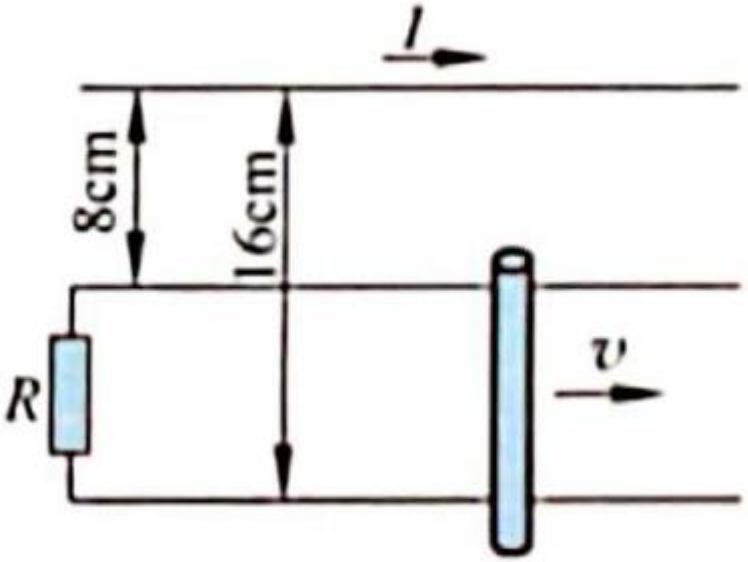


Fig. 8-26 Problem 13.

14. Consider a 6.0V battery attached to two conducting frictionless rails 0.10m apart. There is a magnetic field B of magnitude 0.30T perpendicular to the rails, and a conducting bar can slide over the rails perpendicular to them as well as to the field (Fig. 8-27). The bar is placed on the rails, starts from rest, and accelerates.

(a) What is the direction of its motion?

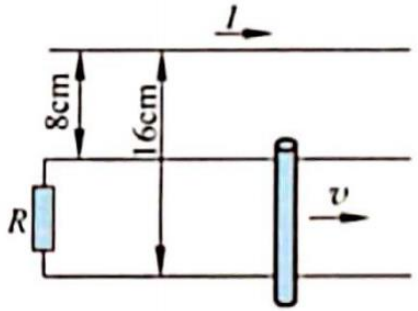


Fig. 8-26 Problem 13.

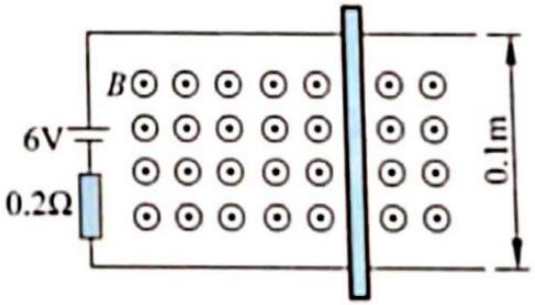


Fig. 8-27 Problem 14.

(b) What is the direction of emf induced?

(c) Given that the total resistance of the closed circuit is 0.20Ω, calculate the current in the bar when its speed is $1.0\text{m}\cdot\text{s}^{-1}$.

Fig. 8-28 Problem 18.

Fig. 8-29 Problem 19.

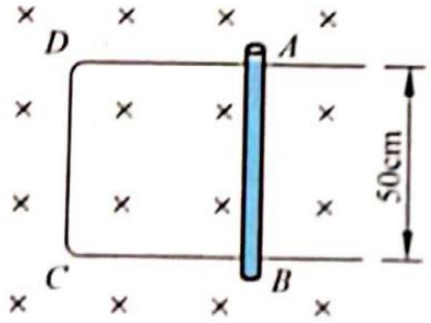


Fig. 8-30 Problem 20.

20. A conducting rod AB in Fig. 8-30 makes contact with the metal rails CB and DA. The apparatus is in a uniform magnetic field 0.5T, perpendicular to the plane of the diagram.

(a) Find the magnitude and direction of the emf induced in the rod when it is moving toward the right with a speed $4\text{m}\cdot\text{s}^{-1}$.

(b) If the resistance of the circuit $ABCD$ is 0.2Ω (assumed constant), find the force required to maintain the rod in motion. Neglect friction.

(c) Compare the rate at which mechanical work is done by the force (Fv) with the rate of development of heat in the circuit (I^2R).